

A high-order meshless departure-point interpolation operator for semi-Lagrangian lattice Boltzmann methods: theory, provenance, and algorithm specification

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Abstract

This report specifies the theoretical basis for replacing the tensor-product Lagrange interpolation of an interpolation-supplemented (semi-Lagrangian) lattice Boltzmann solver with an operator constructed by the Local Anisotropic Basis Function Method (LABFM) of King, Lind and Nasar [1, 2]. The central observation is that the LABFM moment matrix is independent of the operator being approximated; consequently a single local factorisation yields, at no additional matrix cost, a family of *off-node interpolation* operators—one per discrete velocity—in addition to the differential operators for which LABFM was originally formulated. Polynomial reproduction, constant preservation, the degenerate zero-displacement limit, the truncation error and its amplification constant, a von Neumann stability criterion for the streaming step, and invariance under local affine rescaling are each established with proof. A generalised moving least squares formulation of the same departure-point operator is specified alongside as a control arm; it is proved to possess identical polynomial reproduction, so that the order of accuracy is *not* a discriminator between the two constructions and the choice between them reduces to a single structural difference in the conditioning of the local system. Every statement is labelled either with an explicit equation-level citation to the source publication, with a proof given here, or as an open item that must be measured before it may be used as an implementation assumption. One correction is load-bearing: the boundary claim of the 2020 paper, that incomplete stencils yield one-sided differential operators of the internal order, is superseded by the 2021 paper, in which the same authors report their tested construction to be unusable above fourth order and unstable at fourth order, and replace it by a finite-difference hybrid. This adverse result transfers as a conditioning warning for a fixed moment matrix, but not as a proof that every differently selected and scaled one-sided interpolation cloud must fail. The present P6/31 selector therefore accepts wall clouds only through explicit local hard gates, while global interpolation stability remains to be measured. A verification ladder is provided whose rungs reproduce published results, so that an implementation can be audited against the primary literature at each stage. The literature does not supply a universal support construction or a three-dimensional critical stencil ratio. Section 3.7 therefore records, separately and explicitly as a project design rather than a literature result, the current two-dimensional physical $P_6/31$ -point stencil selector, including its mandatory candidate sets, rank and condition-number gates, and compact-first repair policy. A future fully three-dimensional selector remains an open item.

Label	Meaning
[1]	King, Lind & Nasar, <i>J. Comput. Phys.</i> 415 (2020) 109549. All section and equation numbers refer to the published version , a copy of which is held in docs/literature/; preprint numbering is not used.
[2]	King & Lind, <i>J. Comput. Phys.</i> 449 (2022) 110760. Same convention. Note that this paper rennumbers the original-ABF definition [1, Eq. (19)] as its own Eq. (8).
[N]	Derived in the present report; a proof or an explicit construction is given.
[U]	Open item. Not established by the literature and not proved here. Must be measured. No such item may be used as an implementation assumption.

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1 Scope and position with respect to the literature

1.1 Scope

The target solver discretises the discrete-velocity Boltzmann equation on a body-fitted curvilinear grid and advances it by an interpolation-supplemented (semi-Lagrangian) scheme: a collision step evaluated pointwise, followed by transport along the exact characteristics of the free-streaming operator, with the distribution function at the departure point recovered by interpolation. In the present solver that interpolation is a tensor-product seven-point Lagrange interpolation performed in *computational* space; because the characteristic is a straight line in physical space but not in computational space, the departure point must be obtained by numerical integration of the contravariant velocity, which in turn requires the metric terms of the mapping and a free-stream preservation constraint on their discretisation.

This report specifies an alternative in which the interpolation is performed directly in *physical* space on a node cloud, using operators constructed by LABFM. The departure point is then available in closed form, and the metric terms together with the characteristic integration are eliminated. The collision model, the forcing scheme, the statistics, the checkpointing and the domain decomposition are unchanged.

The concrete selector presently implemented and audited in `Lagrange_vs_LABFM/2.labfm/` reconstructs in the physical (y, z) plane: it uses the two-dimensional degree-six space ($p = 27$ nonconstant moments), thirty physical neighbours and a structural centre weight. Section 3.7 is the normative description of that implementation. Statements below about an 83-dimensional three-dimensional basis describe the prospective generalisation, not the current P6/31-point code path.

What is *not* eliminated. An earlier draft of this report also claimed that the solver-wide near-wall ghost-layer infrastructure would be removed, on the strength of the abstract of [1] (“At domain boundaries (with incomplete support) LABFM automatically provides one-sided differences of the same order as the internal scheme, up to 4th order”). **That blanket claim is withdrawn.** [2, §7.1] instead reports failure or instability for its one-sided differential construction and adopts a finite-difference hybrid. The current LABFM selector itself reads physical nodes only and clamps its cloud into the domain, but the existing ghosts remain required by the wall boundary, incumbent ITB/GILBM paths and statistics contract. Local P6/31 solve gates do not establish global wall stability. See §§3.7, 3.10 and open item U4.

1.2 Why the operator route is not pursued

The alternative of discretising $\partial_t f_q + \mathbf{e}_q \cdot \nabla f_q = \Omega_q$ directly, using LABFM *differential* operators and an explicit Runge–Kutta integrator, is not pursued here, for two reasons. First, it has been realised with radial basis function-generated finite differences [15]. Second, and decisively, the convective operators of LABFM are shown in [1, §4.2] to possess eigenvalues with strictly positive real part on any node distribution carrying disorder, for every fundamental radial basis function tested; purely convective problems are therefore unconditionally unstable and require hyperviscosity [1, Eq. (25)]. The same conclusion is reached independently in [4, Fig. 10]. For a wall-resolved simulation of separated turbulence, artificial dissipation of the magnitude required would contaminate the wall shear stress, which is the primary quantity of interest.

The semi-Lagrangian route avoids this failure mode entirely, because transport is effected by translation along exact characteristics rather than by a discrete gradient. The relevant stability question becomes that of the interpolation operator, which is addressed in §4.4.

1.3 Prior art

Meshless lattice Boltzmann formulations date from [10]. Semi-Lagrangian streaming with local radial basis function interpolation was introduced in [11], and with moving least squares reconstruction in [14]. The convergence behaviour of the meshless lattice Boltzmann method has been characterised in [12, 13], where a saturation law is established: the error of the meshless scheme at a given streaming distance is bounded below by the error of the standard lattice Boltzmann method at that same streaming distance, so that node refinement ceases to reduce the error once the interpolation contribution falls below that floor. Separately, high-order semi-Lagrangian kinetic schemes on body-fitted element meshes have been advanced to three-dimensional wall-bounded turbulence [16, 17, 18, 19], and interpolation-based lattice Boltzmann methods have been applied to direct numerical simulation of turbulent channel and duct flows [20].

Against this background, the contribution specified here is narrow and is stated precisely: (i) LABFM has not previously been coupled to a kinetic scheme; (ii) LABFM has been formulated only for differential operators, and its extension to off-node interpolation is given in §4 of this report; and (iii) the polynomial consistency *realised* here exceeds that reported for any existing meshless lattice Boltzmann formulation, in three dimensions and for wall-resolved separated turbulence. Item (iii) is deliberately a statement about realisation and not about capability: Theorem 5 establishes that the same consistency order is attainable by moving least squares, so high order on a node cloud is not specific to LABFM. Claims of novelty beyond these three are not supported by the literature survey and are not made.

2 Notation

The domain is discretised by a node cloud. Node i is at $\mathbf{r}_i$; the difference of any quantity between nodes is written $(\cdot)_{ji} = (\cdot)_j - (\cdot)_i$ and $\mathbf{r}_{ji} = \mathbf{r}_j - \mathbf{r}_i$. Following [2], s_i denotes the mean node spacing in the neighbourhood of node i and h_i the stencil length scale; the support radius is $2h_i$ throughout [1, §2], so that the computational stencil is $\mathcal{N}_i = \{j : |\mathbf{r}_{ji}| \leq 2h_i\}$ with cardinality N_i . The order of polynomial reproduction is m (denoted k in [1] and m in [2]; the latter convention is used here). The number of spatial dimensions is μ . Lattice quantities follow the usual D3Q19 convention, with $\mathbf{e}_q$, $q = 0, \dots, 18$, the discrete velocities and Δt the global time step.

Convention: “order” is not used as a primary index. The word is read differently in the two settings this report has to bridge, and the difference is exactly one. For a *derivative* operator, reproduction of $\mathbb{P}_m$ gives m -th order accuracy, and [1, 2] accordingly write “ m -th order” for degree m ; their statement that one-sided LABFM is unstable “at fourth order” means at $m = 4$. For an *interpolation* operator, reproduction of $\mathbb{P}_m$ gives truncation $O(h^{m+1})$, so the same degree is conventionally called $(m+1)$ -th order. A seven-point one-dimensional Lagrange interpolant is therefore “degree six” under the first reading and “seventh order” under the second, and both descriptions are correct.

To remove the ambiguity, **every statement in this report is indexed by the polynomial degree m , with the truncation exponent written explicitly where it matters.** Ordinal words are used only when quoting a source, and are then tied to the degree. The incumbent scheme of the target solver is a seven-point tensor Lagrange interpolant, i.e. $m = 6$, truncation $O(h^7)$.

The basis dimension follows from m and μ alone. For interpolation in the streamwise–wall-normal plane the relevant counts are

1-D points	degree m	truncation	$\dim \mathbb{P}_m^{2D}$	in-plane neighbours p_2	tensor points	in-plane
5	4	$O(h^5)$	15	14	25	
6	5	$O(h^6)$	21	20	36	
7 (incumbent)	6	$O(h^7)$	28	27	49	
8	7	$O(h^8)$	36	35	64	

$p_2 = \dim \mathbb{P}_m^{2D} - 1$, the constant being carried structurally by the increment form and not by a neighbour.

3 The local anisotropic basis function method

This section is a review. Every equation carries an explicit citation to its source; Appendix A collects the correspondence in tabular form.

3.1 Discrete operators and the moment system

Define the vector of monomials and the vector of partial derivative operators [1, Eqs. (1),(2)], [2, Eqs. (1),(2)]

$$\mathbf{X}_{ji} = \left[x_{ji}, y_{ji}, \frac{x_{ji}^2}{2}, x_{ji}y_{ji}, \frac{y_{ji}^2}{2}, \frac{x_{ji}^3}{6}, \dots, \frac{y_{ji}^m}{m!} \right]^\top, \quad (1)$$

$$\mathbf{D}(\cdot) = [\partial_x, \partial_y, \partial_{xx}, \partial_{xy}, \partial_{yy}, \partial_{xxx}, \dots, \partial_y^m]^\top(\cdot). \quad (2)$$

It is emphasised, as in [1], that $\mathbf{X}_{ji} \neq \mathbf{X}_j - \mathbf{X}_i$: the entries of $\mathbf{X}_{ji}$ are monomials in the separation vector $\mathbf{r}_{ji}$. With this convention the multidimensional Taylor expansion takes the compact form [1, Eq. (3)]

$$(\cdot)_j = (\cdot)_i + \mathbf{X}_{ji} \cdot \mathbf{D}(\cdot)|_i. \quad (3)$$

A general discrete operator is written [1, Eq. (4)], [2, Eq. (3)]

$$\mathcal{L}_i^d(\cdot) = \sum_{j \in \mathcal{N}_i} (\cdot)_{ji} w_{ji}^d \approx \mathbf{C}^d \cdot \mathbf{D}(\cdot)|_i, \quad (4)$$

in which $\mathbf{C}^d$ selects the derivative combination to be approximated [1, Eq. (5)], [2, Eq. (4)]; for the gradient components and the Laplacian, $\mathbf{C}^x = [1, 0, 0, \dots]^\top$, $\mathbf{C}^y = [0, 1, 0, \dots]^\top$ and $\mathbf{C}^L = [0, 0, 1, 0, 1, 0, \dots]^\top$.

The weights are expanded in a set of anisotropic basis functions (ABFs) $\mathbf{W}_{ji} = \mathbf{W}(\mathbf{r}_{ji}/h_i)$ [1, Eq. (10)], [2, Eq. (5)],

$$w_{ji}^d = \mathbf{W}_{ji} \cdot \boldsymbol{\Psi}_i^d, \quad (5)$$

which upon substitution into the moment vector $\mathbf{B}_i^d = \sum_j \mathbf{X}_{ji} w_{ji}^d$ [1, Eq. (8)] yields the local linear system [1, Eqs. (13),(14)], [2, Eqs. (6),(7)]

$$\mathbf{M}_i = \sum_{j \in \mathcal{N}_i} \mathbf{X}_{ji} \otimes \mathbf{W}_{ji}, \quad \mathbf{M}_i \boldsymbol{\Psi}_i^d = \mathbf{C}^d. \quad (6)$$

Remark 1 (Operator independence of the moment matrix). *The matrix $\mathbf{M}_i$ does not depend on d . This is stated explicitly in [1, §3.1]: “The matrix $\mathbf{M}_i$ is the same for all operators (different d), and so we calculate each $\mathbf{M}_i$ once, then solve (14) for each d to obtain the required operators.” The construction of §4 rests entirely on this property: it replaces $\mathbf{C}^d$ by a family of right-hand sides indexed by the discrete velocity, so that one factorisation per node serves all streaming directions as well as every post-processing differential operator.*

Remark 2 (Distributional reading: what $\mathbf{C}$ is; [N]). Associate with the weights the discrete measure

$$\mu_i^d \equiv \sum_{j \in \mathcal{N}_i} w_{ji}^d (\delta_{\mathbf{r}_j} - \delta_{\mathbf{r}_i}), \quad \text{so that} \quad \langle \mu_i^d, \phi \rangle = \mathcal{L}_i^d(\phi). \quad (7)$$

The moment conditions (6) then read: μ_i^d agrees with a prescribed target distribution when paired with every element of the test space $\mathbb{P}_m$, and $\mathbf{C}^d$ is precisely the vector of moments of that target, $\mathbf{C}^{d,\beta} = \langle \text{target}, X^\beta \rangle$. Explicitly,

Target distribution	Resulting $\mathbf{C}$
$-\partial_x \delta_{\mathbf{r}_i}$	$\mathbf{C}^x = [1, 0, 0, \dots]^\top$
$\nabla^2 \delta_{\mathbf{r}_i}$	$\mathbf{C}^L = [0, 0, 1, 0, 1, 0, \dots]^\top$
$\delta_{\mathbf{r}_i + \delta_q} - \delta_{\mathbf{r}_i}$	$\mathbf{C}^q = \mathbf{X}(\delta_q) \quad (\S 4)$

The last line follows from $\langle \delta_{\mathbf{r}_i + \delta_q} - \delta_{\mathbf{r}_i}, X^\beta(\cdot - \mathbf{r}_i) \rangle = X^\beta(\delta_q) - X^\beta(\mathbf{0}) = X^\beta(\delta_q)$, and incidentally explains why $\mathbf{X}$ carries no constant term: the difference form annihilates constants on both sides. This reading unifies the operators of [1, 2] with the one introduced in §4 — they differ only in the target distribution.

3.2 Order of accuracy and basis dimension

When (6) is solved with $\mathbf{M}_i$ of size $p \times p$, exact reproduction of polynomials of total degree m is enforced and the leading truncation error of $\mathcal{L}_i^d$ is $O(h_i^{m+1-l})$, where l is the order of the derivative approximated [1, §3.1], [2, §2]. Accordingly the gradient converges at order m , the Laplacian at order $m - 1$, and — of direct relevance here — an operator with $l = 0$, that is an interpolation, converges at order $m + 1$.

The basis dimension is [1, Eqs. (15),(16)]

$$p_{2D} = \frac{m^2 + 3m}{2}, \quad p_{3D} = \sum_{q=2}^{m+1} \frac{q(q+1)}{2} = \binom{m+3}{3} - 1. \quad (8)$$

m	2	4	5	6	8	10
p_{2D}	5	14	20	27	44	65
p_{3D}	9	34	55	83	164	285
interpolation order $m + 1$	3	5	6	7	9	11

The three-dimensional values reproduce the sequence 3, 9, 19, 34, 55 quoted in [1, Eq. (16)] for $m = 1, \dots, 5$. The polynomial design point is $m = 6$ (truncation $O(h^7)$), matching the degree of the seven-point Lagrange interpolation it replaces. The current in-plane implementation has $p_{2D} = 27$ and is specified in §3.7; a genuinely three-dimensional implementation would instead have $p_{3D} = 83$ and is not specified by the present source files.

3.3 Structure of the weight system: what the basis functions decide

The material of this subsection is not stated in this form in [1, 2], but it is the shortest route to understanding both the role of the ABFs and the relationship to the control formulation of §5. [N]

Collect the stencil data into matrices: let $\mathbf{X} \in \mathbb{R}^{N_i \times p}$ have rows $\widetilde{\mathbf{X}}_{ji}^\top$ and $\mathbf{W} \in \mathbb{R}^{N_i \times p}$ have rows $\mathbf{W}_{ji}^\top$, and let $\mathbf{w} \in \mathbb{R}^{N_i}$ collect the weights. The moment conditions (6) are

$$\mathbf{X}^\top \mathbf{w} = \mathbf{C} \quad \text{— } p \text{ equations in } N_i \text{ unknowns.} \quad (9)$$

For $N_i > p$ this system is underdetermined: its solution set is an affine subspace of dimension $N_i - p$, and *every member reproduces $\mathbb{P}_m$ exactly*. A selection rule is therefore mandatory. The ABF ansatz (5) supplies one by restricting $\mathbf{w}$ to the column space of $\mathbf{W}$, which renders the system square, and $\mathbf{M}_i = \mathbf{X}^\top \mathbf{W}$ is exactly that restriction. Writing the rule with a general *selector* $\mathbf{S} \in \mathbb{R}^{N_i \times p}$,

$$\boxed{\mathbf{w} = \mathbf{S}(\mathbf{X}^\top \mathbf{S})^{-1} \mathbf{C}} \quad \begin{cases} \mathbf{S} = \mathbf{W} & \text{LABFM,} \\ \mathbf{S} = \mathbf{\Theta} \mathbf{Q} & \text{generalised moving least squares (§5).} \end{cases} \quad (10)$$

Proposition 1 (Degenerate limit $N_i = p$; [N]). *If $N_i = p$ and both $\mathbf{X}$ and $\mathbf{S}$ are invertible, then $\mathbf{w} = \mathbf{X}^{-\top} \mathbf{C}$, independent of $\mathbf{S}$. The weights are then those of plain polynomial interpolation of degree m ; in one dimension with m neighbours plus the centre they are the Lagrange weights on $m + 1$ points.*

Proof. $\mathbf{w} = \mathbf{S}(\mathbf{X}^\top \mathbf{S})^{-1} \mathbf{C} = \mathbf{S} \mathbf{S}^{-1} \mathbf{X}^{-\top} \mathbf{C} = \mathbf{X}^{-\top} \mathbf{C}$. Uniqueness of the solution of the square system (9) identifies it with the unique degree- m polynomial reproduction rule on those $N_i + 1$ points. $\square$

Remark 3 (What the surplus degrees of freedom buy). *Proposition 1 shows that at $N_i = p$ the basis functions are irrelevant: they cancel. They begin to act only for $N_i > p$, and by (9) every admissible choice has identical polynomial consistency. The surplus $N_i - p$ degrees of freedom therefore cannot change the order of accuracy; what they do change is the coefficient of the leading error term, together with the conditioning and stability of the resulting operator (Appendix B exhibits both statements explicitly). This is consistent with the requirement $N_{\text{crit}} > N_{\text{poly}}$ reported in [1, §3.2] — for instance 37 against 28 at $m = 6$ in two dimensions — and it is the precise sense in which LABFM and moving least squares differ: same constraint set, different selector.*

Proposition 2 (The expansion origin is a gauge; [N]). *The constraint (9) is invariant under a change of the point about which the monomials $\mathbf{X}$ and the right-hand side $\mathbf{C}$ are formed. Consequently, with $\mathbf{S}$ and N_i held fixed, the weights $\mathbf{w}$ do not depend on that choice.*

Proof. The constraint is equivalent to the origin-free statement “ $\sum_j w_{ji}(P_j - P_i) = \langle \text{target}, P \rangle$ for all $P \in \mathbb{P}_m$ ”. Monomials about any point form a basis of $\mathbb{P}_m$ (the constant contributing the trivial identity $0 = 0$ in the difference form), so different choices produce different but equivalent bases for the same p conditions, hence the same solution set; intersecting with the fixed subspace $\text{span}(\mathbf{S})$ gives the same $\mathbf{w}$. $\square$

Remark 4 (Three distinct “origins”, only one of which is free). *Formulae in [1] place three things at $\mathbf{r}_i$, and they are commonly conflated: (i) the monomial/Taylor origin, which is a gauge by Proposition 2; (ii) the ABF argument origin, which is not a gauge, since it defines $\text{span}(\mathbf{S})$ and hence which admissible solution is selected, together with κ , Λ and $|G_q|$; and (iii) the stencil centre, which is not a gauge either, since it determines N_i . Gauge freedom in (i) is nevertheless not licence: with the origin far from the stencil the entries of $\mathbf{X}$ span many orders of magnitude and $\mathbf{M}_i$ becomes numerically intractable although formally equivalent. Choosing $\mathbf{r}_i$, together with the scaling of §4.5, is the well-conditioned gauge. Proposition 2 is used as a verification test in rung L5f of §6. The alternative of centring (ii) and (iii) on the departure point is a genuine method variant, adopted for example in the meshless lattice Boltzmann scheme of [12]; it costs one moment matrix per discrete velocity.*

3.4 Construction of the anisotropic basis functions

Two constructions appear in the literature.

Original ABFs. In [1] the ABFs are generated as partial derivatives of a fundamental radial basis function W^0 ,

$$\mathbf{W}_{ji} = \mathbf{D} W^0(\mathbf{r}_{ji}), \quad (11)$$

[1, Eq. (19)], with four choices of W^0 examined (conic, quadratic, Wendland C6, Gaussian; explicit forms in [1, Appendix A]). The fundamental functions are normalised so that $W^0(2) = 0$ (Gaussian excepted) and $2\pi \int_0^2 q W^0(q) dq = 1$ [1, Eq. (A.1)]; the convex quadratic is $W_Q(q) = \frac{3}{16\pi}(q-2)^2$ on $0 \leq q \leq 2$ [1, Eq. (A.3)], designed so that $dW_Q/dq|_{q=2} = 0$, which [1] identifies as apparently necessary for stability of the convective operator on a uniform distribution.

Remark 5 (Implementation caveat required to reproduce [1]; [N]). *The full expansion [1, Eq. (A.2)] of $\mathbf{D}W^0$ contains terms in d^2W^0/dr^2 and higher radial derivatives, which vanish only for the conic W^0 . [1, Appendix A] states that in practice these terms are omitted for all $m \geq 4$ (that is, from the tenth element of $\mathbf{W}_{ji}$ onward) regardless of whether they vanish, for two stated reasons: computational simplicity, and greater resilience to node-distribution disorder. Any attempt to reproduce the published two-dimensional convergence and critical-ratio results — rungs L1 and L2 of §6 — must apply the same truncation, or the numbers will not match. This caveat does not affect the Hermite construction, which is what the production path uses.*

Hermite ABFs. In [2] the ABFs are instead built from an orthogonal polynomial basis. Index the basis by the multi-index α of the corresponding monomial: if the α -th entry of $\mathbf{X}_{ji}$ is proportional to $x_{ji}^a y_{ji}^b$, then [2, Eqs. (10),(11)]

$$W_{ji}^\alpha = \frac{\psi(|\mathbf{r}_{ji}|/h_i)}{\sqrt{2^{a+b}}} H_a\left(\frac{x_{ji}}{h_i\sqrt{2}}\right) H_b\left(\frac{y_{ji}}{h_i\sqrt{2}}\right), \quad (12)$$

where H_a is the physicists' Hermite polynomial of degree a and ψ is a radial basis function (Wendland C2 by default; [2] states that other choices, including $\psi \equiv 1$, are admissible).

Remark 6 (The roles of the two subscripts, and the shape of $\mathbf{W}$). W_{ji}^α carries three indices: α selects the basis function, j the neighbour at which it is evaluated, and i the centre. The object $\mathbf{W}_{ji} = [W_{ji}^1, \dots, W_{ji}^p]^\top$ appearing in (5) is therefore, for one fixed neighbour, the vector of all p basis functions evaluated at the single separation $\mathbf{r}_{ji}$. Every neighbour carries every basis function; the basis is not distributed among the neighbours.

Two consequences are worth stating, because the alternative reading is a common one. First, dimensionally: only if $\mathbf{W}_{ji}$ is a p -vector is $\mathbf{X}_{ji} \otimes \mathbf{W}_{ji}$ a $p \times p$ matrix and $\mathbf{M}_i$ a square system; were each neighbour to carry a single basis function, the sum in (6) would return a vector and the construction would not close. Equivalently, in the notation of §3.3, $\mathbf{W}$ is an $N_i \times p$ matrix with neighbours indexing rows and basis functions indexing columns — the same shape as $\mathbf{X}$, and the same shape as a design matrix in least squares.

Second, by contrast: radial basis function methods do index the basis by node, placing one function centred at each neighbour, which yields an $N_i \times N_i$ system whose size grows with the stencil. In LABFM the basis is indexed by differential order instead, so enlarging the stencil adds equations to an overdetermined set rather than unknowns, and the local system remains $p \times p$ whatever N_i is. This is the origin of the $O(p^3)$ rather than $O(N_i^3)$ factorisation cost, and of the surplus degrees of freedom discussed in Remark 3.

Decisive result for the three-dimensional case. [2, §3] reports: “An example of where this results in the method breaking down is when LABFM is extended to three-dimensions, with $m = 4$. Here M has size $n = 34$, and rank $n - 1 = 33$, *regardless of the node distribution*.” The original ABFs (11) therefore render M rank deficient in three dimensions for $m \geq 4$, independently of the node cloud. The present application is three-dimensional with target $m = 6$; the Hermite construction (12) is consequently *mandatory*. The original construction retains value only as a two-dimensional regression target (rungs L1–L2 of §6) and as the subject of the negative test L3, and must not appear on any three-dimensional production path.

3.5 Row preconditioning

Define the vector of inverse powers of h_i [2, Eq. (9)]

$$\mathbf{H}_i = [h_i^{-1}, h_i^{-1}, h_i^{-2}, h_i^{-2}, h_i^{-2}, h_i^{-3}, \dots, h_i^{-m}]^\top, \quad \widetilde{\mathbf{X}}_{ji} = \mathbf{X}_{ji} \cdot \mathbf{H}_i, \quad \widetilde{\mathbf{C}}^d = \mathbf{C}^d \cdot \mathbf{H}_i, \quad (13)$$

the products being taken entrywise. Solving (6) with $\widetilde{\mathbf{X}}$ and $\widetilde{\mathbf{C}}$ in place of $\mathbf{X}$ and $\mathbf{C}$ leaves the exact solution unchanged while improving the condition number of M by several orders of magnitude [2, §2]. This preconditioning is applied unconditionally in the present specification.

3.6 Sampling requirements and stencil sizing

The construction requires that the discrete sums forming M_i approximate the corresponding continuous integrals [1, Eq. (22)]. This imposes a lower bound on h/s , made precise in [1, §3.2] by an angular sampling argument: expressed in local polar coordinates, the order- m ABFs contain terms in $\cos m\theta$ and $\sin m\theta$, so that if the support is divided into $2m$ equal angular sectors, each sector must contain at least one node. Below the resulting critical ratio, some eigenvalues of M_i fall to machine-precision zero and the condition number increases by roughly ten orders of magnitude [1, §3.2].

Remark 7 (Normalisation of the sampling condition; [N]). *The inner product of [1, Eq. (18)] is a mean over the support, $\langle f, g \rangle = V_i^{-1} \int_{V_i} f g \, dV_j$, so that [1, Eq. (22)] as printed, $\sum_j X_{ji}^a W_{ji}^b \approx \langle X^a, W^b \rangle$, is short of a factor N_i . With node number density $s^{-\mu}$ the operative relation is*

$$\sum_{j \in \mathcal{N}_i} X_{ji}^a W_{ji}^b \approx \frac{1}{s_i^\mu} \int_{V_i} X^a W^b \, dV = N_i \langle X^a, W^b \rangle. \quad (14)$$

The distinction is immaterial to the argument of [1] but must be respected when a continuum reference matrix is computed as a diagnostic (§3.8).

Published two-dimensional values. These belong to *two different* ABF constructions and must not be conflated:

Quantity	ABF used	$m=2$	4	6	8	10
$(h/s)_{\text{crit}}$, <i>critical</i> [1, §3.2]	original	0.75	1.15	1.60	2.10	—
N_{crit} [1, §3.2]	original	8	21	37	57	—
$N_{\text{poly}} = p_{2D} + 1$	—	6	15	28	45	66
h/s , <i>recommended</i> [2, §5]	Hermite	1.2	1.4	1.8	2.3	2.8

The stencil optimisation of [2, §4] reduces the mean ratio for $m = 6$ from 2.2 to approximately 1.5, lowering the neighbour count by 17.5%.

[U] Open item U8. No *critical* ratio is published for the Hermite ABFs, in any dimension. The critical values above were measured with the original ABFs, which the production path does not use; the Hermite values quoted by [2] are recommended operating points carrying an unquantified margin. Code must therefore not name the original-ABF critical values as if they applied to the production construction.

The neighbour count follows from the support geometry. In two dimensions $N_i \approx 4\pi(h_i/s_i)^2$ [2, §2]; the same argument in three dimensions gives [N]

$$N_i^{3D} \approx \frac{4}{3}\pi \frac{(2h_i)^3}{s_i^3} = \frac{32\pi}{3} \left(\frac{h_i}{s_i}\right)^3 \approx 33.51 \left(\frac{h_i}{s_i}\right)^3. \quad (15)$$

[U] Open item U1. The critical ratio $(h/s)_{\text{crit}}^{3D}$ is not available. The values tabulated above are two-dimensional. The three-dimensional sampling condition is a solid-angle rather than a plane-angle condition and is necessarily more demanding, but by how much is not established; [2] states explicitly that three-dimensional simulations were reserved for a later study. Because N_i scales as $(h/s)^3$, this single unknown controls any cost model built on the neighbour count.

3.7 Current project-specific $P_6/31$ -point stencil selector

This subsection is an implementation specification, not a rule taken from [1, 2]. Its normative sources are `Lagrange_vs_LABFM/2.labfm/labfm_stencil.h` and `Lagrange_vs_LABFM/2.labfm/labfm_solve.h`, read through the active call chain `b_weights_case2` → `labfm_select_physical_stencil` → `labfm_candidate_pool_build`. The separately named `labfm_select_physical_stencil_legacy_rrqr` has no caller in this path and is retained only for validation; it does not define the current selector. The Taylor, Hermite and Wendland functions called by this path are defined in `Lagrange_vs_LABFM/2.labfm/labfm_hermite.h`.

Project rule (a posteriori certificate, not an a priori topology theorem). The following five stages map one-to-one to §§3.7.1–3.7.5:

1. **Point sets.** Construct a deterministic physical-node candidate pool and distinguish points protected in that pool from points selected in the final 31-point footprint.
2. **Compact tier.** First try the fixed seven-layer index-space pyramid. Its actual physical shape may be warped, and the template is not assumed to be nonsingular before it is tested.
3. **Admissibility certificate.** Assemble the matrices from the actual physical coordinates. Accept a trial only when its support, numerical rank, reciprocal conditioning and direction-independent weight-amplification gates all pass.
4. **Repair tier.** Only if every compact-tier trial fails, generate replacement clouds, apply the same admissibility certificate, and compare the surviving clouds by a degree-seven error proxy.
5. **Final solve and export.** For each lattice direction, solve for its weights and pass independent solver and backward-error checks. Failed data are never exported to the time-step path.

Thus geometry proposes a stencil, algebra certifies it, and only a certified stencil may be used. Conditioning is an acceptance gate, not the optimisation objective.

3.7.1 From grid indices to three point sets

Purpose of this stage. This stage answers three different questions: which physical nodes are enumerated, which of them are guaranteed a place in the finite candidate pool, and which 30 neighbours finally accompany the centre. These three sets must not be conflated.

Geometric notation. The code stores three ghost layers around the physical array. Let (j, k) be the stored array indices of the arrival node at which a stencil is being built. Its physical-grid indices are

$$c_j = j - 3, \quad c_k = k - 3. \quad (16)$$

Here N_y and N_z are, symmetrically, the numbers of distinct physical-grid nodes in the streamwise and wall-normal directions. Their canonical physical indices are $0, \dots, N_y - 1$ and $0, \dots, N_z - 1$, respectively. The streamwise coordinate file contains one additional record at index N_y that repeats index zero at the other end of the period. After retaining this repeated endpoint and adding three ghosts on each side, the stored extents are

$$N_{y6} = (N_y + 1) + 6 = N_y + 7, \quad N_{z6} = N_z + 6. \quad (17)$$

Thus the implementation variables satisfy $\text{NI} = N_y + 1$ (the solver macro NY has the same convention), $\text{NJ} = N_z$, $\text{NY6} = N_{y6}$, and $\text{NZ6} = N_{z6}$; N_{z6} is the stored wall-normal stride.

For one enumerated candidate, the following symbols are used:

Symbol	Meaning
d_j, d_k	Integer offsets from (c_j, c_k) in structured index space. They determine the candidate through $c_j + d_j = j_p + \sigma_p N_y$ and $k_p = c_k + d_k$; d_j is kept unwrapped so that its periodic image remains known.
j_p, k_p	Canonical physical-grid indices of the candidate, with $0 \leq j_p < N_y$ and $0 \leq k_p < N_z$.
ι_p	Canonical physical-node label, stored as <code>physical_id</code> ; it uniquely identifies (j_p, k_p) but does not distinguish its periodic images.
σ_p	Integer periodic-image number, stored as <code>image_shift</code> and uniquely defined by $c_j + d_j = j_p + \sigma_p N_y$. It identifies which periodic image gives a continuous streamwise displacement.
L_y	Physical streamwise period, stored as <code>period_y</code> .
Y, Z	Physical-coordinate arrays; $Y(j_p, k_p)$ and $Z(j_p, k_p)$ are their values at the canonical physical node.
$\Delta Y_p, \Delta Z_p$	Physical-coordinate displacement from the arrival node to the chosen periodic image of the candidate.

If (Y_0, Z_0) denotes the physical coordinate of the arrival node, then

$$\Delta Y_p = Y(j_p, k_p) + \sigma_p L_y - Y_0, \quad \Delta Z_p = Z(j_p, k_p) - Z_0. \quad (18)$$

The canonical physical-node label used for de-duplication is

$$\iota_p = j_p N_z + k_p, \quad 0 \leq \iota_p < N_y N_z. \quad (19)$$

The code stores ι_p in `physical_id`. It is not the same as the flattened storage address `storage_id` = $(j_p + 3)N_{z6} + (k_p + 3)$, where N_{z6} is the stored wall-normal stride including ghosts. Two periodic images can have the same ι_p and canonical storage address but different values of σ_p , and hence different continuous coordinates in (18).

Set 1: all enumerated physical candidates. The search is written symmetrically in terms of the two integer offsets:

$$-16 \leq d_j \leq 16, \quad \max(-16, -c_k) \leq d_k \leq \min(16, N_z - 1 - c_k). \quad (20)$$

For every pair (d_j, d_k) , the canonical physical indices are obtained from $c_j + d_j = j_p + \sigma_p N_y$ and $k_p = c_k + d_k$. The first relation wraps the periodic streamwise index while retaining its image shift; the bounds on d_k ensure directly that $0 \leq k_p < N_z$, so no wall ghost can enter the set. The centre $(d_j, d_k) = (0, 0)$ is skipped. Away from both walls the set therefore contains $33 \times 33 - 1 = 1088$

candidates; wall clipping reduces this number. The value 16 is a finite structured-index search budget; it is not the Wendland support radius and is not a physical-distance cutoff.

To order the candidates, the code first estimates the two local physical spacings

$$\begin{aligned} s_y &= \frac{1}{2} (|Y_{j+1,k}^{\text{st}} - Y_{j,k}^{\text{st}}| + |Y_{j,k}^{\text{st}} - Y_{j-1,k}^{\text{st}}|), \\ s_z &= \frac{1}{2} (|Z_{j,k+1}^{\text{st}} - Z_{j,k}^{\text{st}}| + |Z_{j,k}^{\text{st}} - Z_{j,k-1}^{\text{st}}|). \end{aligned} \quad (21)$$

The superscript *st* indicates that these four differences use stored-array indices; the three ghost layers make all four accesses valid even at a physical wall. For each candidate it then defines

$$\rho_p = \sqrt{(\Delta Y_p/s_y)^2 + (\Delta Z_p/s_z)^2}, \quad \theta_p = \text{atan2}(\Delta Z_p, \Delta Y_p) \in [-\pi, \pi]. \quad (22)$$

Here ρ_p is a dimensionless local-grid distance, whereas θ_p is used only to break equal- ρ_p ties. The enumerated array is sorted lexicographically by

$$(\rho_p, \theta_p, \iota_p, \sigma_p). \quad (23)$$

Floating-point keys are compared directly; there is no epsilon-equality rule.

Set 2: the candidate pool of at most 160 points. **Key point:** For an interior arrival node on the audited grids, Set 2 is not the 160 globally nearest points in Set 1: it first retains the 48 neighbours in a protected seven-by-seven index-space window, then retains the 112 nearest unique physical nodes among the other 1040 candidates, where “nearest” is determined first by ρ_p and the remaining keys only break exact ties. Thus

$$\underbrace{1088}_{\text{Set 1}} = \underbrace{48}_{\text{protected}} + \underbrace{1040}_{\text{other candidates}}, \quad \underbrace{160}_{\text{Set 2}} = \underbrace{48}_{\text{protected}} + \underbrace{112}_{\text{nearest retained}}. \quad (24)$$

Pass 0: retain the 48 protected neighbours. The protected index-space window is

$$\begin{aligned} k_{\text{lo}} &= \min(\max(c_k - 3, 0), N_z - 7), \\ \mathcal{W}_{\text{prot}} &= \{(d_j, d_k) : -3 \leq d_j \leq 3, k_{\text{lo}} - c_k \leq d_k \leq k_{\text{lo}} + 6 - c_k\} \setminus \{(0, 0)\}. \end{aligned} \quad (25)$$

Here k_{lo} is the first physical k -index of the protected seven-row block. Away from a wall, $k_{\text{lo}} = c_k - 3$ and hence $-3 \leq d_k \leq 3$. At the lower wall $c_k = 0$, the range becomes $0 \leq d_k \leq 6$; at the upper wall $c_k = N_z - 1$, it becomes $-6 \leq d_k \leq 0$. Thus the same seven-by-seven offset window is shifted inward near a wall without admitting ghosts. Removing $(0, 0)$ excludes the centre, leaving $7 \times 7 - 1 = 48$ protected neighbours on the audited grids. These 48 points enter Set 2 even if some unprotected candidates have smaller ρ_p .

Pass 1: fill the remaining 112 slots by normalised physical distance. For an interior node, Pass 1 scans the other $1088 - 48 = 1040$ candidates in the order $(\rho_p, \theta_p, \iota_p, \sigma_p)$ and inserts the first 112 unique physical nodes. Before each insertion, the pool is searched for the same ι_p (`physical_id` in the code); a later periodic image of an already retained node is discarded. Equivalently, the complete priority key is

$$(\text{pass}, \rho_p, \theta_p, \iota_p, \sigma_p),$$

with protected-window points assigned to Pass 0. The four candidate keys do not receive separate point quotas: ρ_p supplies the primary ordering, while θ_p , ι_p , and σ_p are consulted successively only when all preceding keys are exactly equal.

Near a wall, Set 1 contains fewer than 1088 candidates, so the pool of unprotected candidates is smaller than 1040; the algorithm itself is unchanged. On the audited PH/nonPH grids, enough unique candidates still exist, so Pass 0 retains 48 protected points and Pass 1 still fills the remaining $160 - 48 = 112$ slots. If Set 1 were exhausted earlier, Set 2 would simply contain fewer than 160 points. If this header is reused with $N_y \leq 32$, an explicit guard comparing ι_p with the centre’s

canonical-node label must be added before making the same uniqueness claim; the present input check alone allows $\text{NI} = N_y + 1 \geq 8$, that is, $N_y \geq 7$.

The protected window is needed because, on a strongly stretched grid, the first 160 points in pure ρ_p order can occupy at most six nearly parallel wall-normal layers. Such a set can make the degree-six sampling problem algebraically deficient. Protection keeps a seven-layer construction available to both the compact and repair tiers. Protection means *must enter the pool*, not *must enter the final stencil*.

Set 3: the final 31-point footprint. The final footprint contains the arrival node, labelled $n = 0$, and exactly 30 physical neighbours, labelled $n = 1, \dots, 30$. Only the 30 neighbours form rows of the algebraic matrices:

$$\mathbf{X}, \mathbf{W} \in \mathbb{R}^{30 \times 27}, \quad \mathbf{M} = \mathbf{X}^\top \mathbf{W} \in \mathbb{R}^{27 \times 27}. \quad (26)$$

The centre is not a thirty-first row. After the 30 neighbour weights are obtained, its weight is set to $w_0 = 1 - \sum_{n=1}^{30} w_n$ so that constants are reproduced exactly.

Where the selector stops	Points forced into the final footprint	What may change
Compact tier	All 30 noncentre points of the fixed pyramid	Only the scale parameters change; no point is replaced.
Repair source A	The same 30 pyramid neighbours	The amplification ceiling is relaxed.
Repair source B	24–27 QRCP anchors from the pyramid	The remaining 6–3 points are replaced by eligible non-template candidates.
Repair source C	27 QRCP anchors from a pool prefix	Three additional eligible non-anchor candidates complete the footprint.

Therefore the protected pool window, the nearest points in ρ , and the final 30-neighbour footprint are different sets. Across repaired nodes there is no universal list of 30 index offsets.

3.7.2 Compact tier: fixed index template, then physical-coordinate test

Purpose of this stage. This stage proposes one compact 31-point topology in structured-index space, maps it to the actual physical coordinates, and accepts the first scale choice that passes the algebraic certificate. The topology proposes; it does not prove solvability.

Step 1: construct the template in index space. The first trial is not obtained by taking the 30 nearest points in physical distance. Instead, `b2_pattern` chooses seven consecutive physical k -index layers. Near a wall the entire seven-layer block is shifted into the physical domain. The resulting layer offsets d_k are sorted by increasing $|d_k|$; at equal distance the negative offset is placed first. The numbers of points assigned to the seven ordered layers are

$$(7, 7, 5, 5, 3, 3, 1), \quad (27)$$

where each layer uses the streamwise offset order $d_j = 0, +1, -1, +2, -2, \dots$. The counts sum to 31 and include the centre $(d_j, d_k) = (0, 0)$ once; removing that centre leaves the required 30 neighbours.

For an interior centre, the seven d_k values are

$$0, -1, +1, -2, +2, -3, +3.$$

At the lower wall they are $0, +1, \dots, +6$, and at the upper wall they are $0, -1, \dots, -6$. Thus the template is fixed in structured index space. After the index offsets are mapped through

$Y(j_p, k_p), Z(j_p, k_p)$, however, the physical cloud can be stretched, sheared or curved; it need not look like a regular pyramid or diamond.

The seven-layer topology is intended to avoid the observed failure in which a nearest-point cloud collapses onto at most six nearly parallel layers. Six such physical layers can lie on the zero set of a nonzero degree-six product polynomial, making the P_6 sampling matrix deficient. The template removes that specific geometry in the intended grids, but it is not an analytic proof that every mapped template is unisolvent.

Step 2: define the anisotropic coordinate scaling. Using the local spacings in (21), define

$$A = \text{clamp}(s_y/s_z, 1/4096, 4096), \quad \beta \in \{0, \frac{1}{2}, 1\}, \quad a_y = A^{\beta/2}, \quad a_z = a_y^{-1}, \quad (28)$$

and, for neighbour n , define the scaled physical displacements

$$\bar{Y}_n = \frac{\Delta Y_n}{a_y}, \quad \bar{Z}_n = \frac{\Delta Z_n}{a_z}, \quad \bar{r}_n = \sqrt{\bar{Y}_n^2 + \bar{Z}_n^2}. \quad (29)$$

The bars denote coordinate scaling; they do not denote an average. Because $a_y a_z = 1$, this is an invertible diagonal change of coordinates and does not change the polynomial space that is reproduced.

Step 3: define h and the Wendland support. For each value of β , the template length scale is the root-mean-square radius of its 30 neighbours:

$$h_{\text{tmpl}} = \left(\frac{1}{30} \sum_{n=1}^{30} \bar{r}_n^2 \right)^{1/2}. \quad (30)$$

Hence h is *not* the minimum grid spacing. The code tries

$$h = m_h h_{\text{tmpl}}, \quad m_h \in \{1, 1.2, 1.5, 2\},$$

in the written order. The dimensionless radial coordinate of neighbour n is

$$q_n = \frac{\bar{r}_n}{h} = \frac{\sqrt{(\Delta Y_n/a_y)^2 + (\Delta Z_n/a_z)^2}}{h}. \quad (31)$$

The radial factor is the compactly supported Wendland C^2 function

$$\psi(q) = \begin{cases} (1 - q/2)^4(1 + 2q), & 0 \leq q < 2, \\ 0, & q \geq 2. \end{cases} \quad (32)$$

All 30 template neighbours must satisfy $q_n < 2$; otherwise at least one row of $\mathbf{W}$ is zero and the trial is rejected.

Step 4: compact-first acceptance. The loop order is $\beta = 0, \frac{1}{2}, 1$, with the four h multipliers tested inside each β . The first pair (β, h) that passes every gate in §3.7.3 is accepted immediately. Therefore the compact tier does not optimise over alternative point sets or search for the smallest condition number. It preserves the index-space template whenever that template is numerically acceptable.

3.7.3 Admissibility: rank, conditioning and amplification

Purpose of this stage. This stage answers the central solvability question. No argument based only on the pyramid shape is used to declare $\mathbf{M}$ nonsingular. Instead, $\mathbf{X}$, $\mathbf{W}$ and $\mathbf{M}$ are assembled from the 30 mapped physical points and certified numerically.

Step 1: define the 27 basis labels. Let

$$\mathcal{A}_6 = \{(a, b) \in \mathbb{N}_0^2 : 1 \leq a + b \leq 6\}. \quad (33)$$

This set contains $2 + 3 + 4 + 5 + 6 + 7 = 27$ exponent pairs. The constant pair $(0, 0)$ is omitted because the centre weight later enforces constant reproduction. We use $\mu = \mu(a, b)$ for a column label of $\mathbf{X}$ and $\alpha = \alpha(u, v)$ for a column label of $\mathbf{W}$, where both (a, b) and (u, v) independently range over $\mathcal{A}_6$. The two labels each have 27 possible values, but μ is not set equal to α . More explicitly, $\mu(a, b)$ and $\alpha(u, v)$ mean “the integer column occupied by this exponent pair” in the fixed code ordering; they are column-address maps, not additional polynomial variables.

For neighbour n , divide the scaled displacements in (29) by h :

$$\xi_n = \frac{\bar{Y}_n}{h} = \frac{\Delta Y_n}{a_y h}, \quad \eta_n = \frac{\bar{Z}_n}{h} = \frac{\Delta Z_n}{a_z h}, \quad q_n = \sqrt{\xi_n^2 + \eta_n^2}. \quad (34)$$

Here q_n is the Wendland radial coordinate of a *stencil point*. It is not an LBM direction label; lattice directions will be denoted by d in §3.7.5.

Step 2: define every entry of $\mathbf{X}$, $\mathbf{W}$ and $\mathbf{M}$. For $\mu = \mu(a, b)$, the Taylor row is

$$X_{n,\mu(a,b)} = \frac{\xi_n^a \eta_n^b}{a! b!}. \quad (35)$$

Let H_r be the physicists’ Hermite polynomial, $H_0(x) = 1$, $H_1(x) = 2x$, and $H_{r+1}(x) = 2xH_r(x) - 2rH_{r-1}(x)$. For $\alpha = \alpha(u, v)$, the Hermite–Wendland row is

$$W_{n,\alpha(u,v)} = \psi(q_n) 2^{-(u+v)/2} H_u\left(\frac{\xi_n}{\sqrt{2}}\right) H_v\left(\frac{\eta_n}{\sqrt{2}}\right), \quad (36)$$

where ψ is given by (32). The moment matrix is

$$M_{\mu\alpha} = \sum_{n=1}^{30} X_{n\mu} W_{n\alpha}, \quad \mathbf{M} = \mathbf{X}^\top \mathbf{W}. \quad (37)$$

This is a sum, not an average; dividing it by 30 would only multiply the whole matrix by a nonzero scalar and would not change its rank.

Why the template does not prove that $\mathbf{M}$ is full rank. Let $\mathbf{x}_n$ and $\mathbf{w}_n$ be row n of $\mathbf{X}$ and $\mathbf{W}$. Then

$$\mathbf{M} = \sum_{n=1}^{30} \mathbf{x}_n^\top \mathbf{w}_n. \quad (38)$$

Each point contributes a matrix of rank at most one. Thirty contributions do not automatically span 27 independent directions, and different cross terms can cancel. Moreover, $\mathbf{w}_n$ is not a positive scalar multiple of $\mathbf{x}_n$. For example,

$$X_{n,\mu(2,0)} = \frac{\xi_n^2}{2}, \quad W_{n,\alpha(2,0)} = \psi(q_n)(\xi_n^2 - 1), \quad W_{n,\alpha(3,0)} = \psi(q_n)(\xi_n^3 - 3\xi_n). \quad (39)$$

Therefore $\mathbf{M}$ is not a positive weighted Gram matrix $\mathbf{X}^\top \mathbf{D} \mathbf{X}$, and positive definiteness cannot be inferred from full column rank of $\mathbf{X}$.

An equivalent exact statement follows from thin QR factorizations

$$\mathbf{X} = \mathbf{Q}_X \mathbf{R}_X, \quad \mathbf{W} = \mathbf{Q}_W \mathbf{R}_W, \quad \mathbf{M} = \mathbf{R}_X^\top (\mathbf{Q}_X^\top \mathbf{Q}_W) \mathbf{R}_W. \quad (40)$$

Here the columns of $\mathbf{Q}_X, \mathbf{Q}_W \in \mathbb{R}^{30 \times 27}$ are orthonormal. If $\mathbf{X}$ and $\mathbf{W}$ have full column rank, the triangular factors $\mathbf{R}_X, \mathbf{R}_W$ are nonsingular, and

$$\text{rank}(\mathbf{M}) = 27 \iff \text{rank}(\mathbf{Q}_X^\top \mathbf{Q}_W) = 27 \iff \text{col}(\mathbf{W}) \cap \ker(\mathbf{X}^\top) = \{\mathbf{0}\}. \quad (41)$$

The fixed pyramid has no proven uniform lower bound for this cross-space pairing on every allowed warped grid. It must therefore be checked after its actual coordinates are known.

Step 3: exact uniqueness condition. For a right-hand side $\mathbf{C} \in \mathbb{R}^{27}$, the coefficient vector $\mathbf{\Psi} \in \mathbb{R}^{27}$ is defined by $\mathbf{M}\mathbf{\Psi} = \mathbf{C}$. Its exact unique-solvability condition is

$$\text{rank}(\mathbf{M}) = \text{rank}([\mathbf{M} \mid \mathbf{C}]) = 27. \quad (42)$$

Because $\mathbf{M}$ is a 27×27 square matrix, $\text{rank}(\mathbf{M}) = 27$ alone implies the augmented-rank equality for every $\mathbf{C}$.

Step 4: numerical-rank gates. For each trial, the decisive question is whether the assembled moment matrix $\mathbf{M} = \mathbf{X}^\top \mathbf{W}$ has numerical rank 27. The code does not apply its QR rank test directly to the raw $\mathbf{M}$. It first performs six fixed row/column max-norm equilibration sweeps and forms

$$\widetilde{\mathbf{M}} = \mathbf{D}_r \mathbf{M} \mathbf{D}_c, \quad \text{rank}(\widetilde{\mathbf{M}}) = \text{rank}(\mathbf{M}), \quad (43)$$

The subscripts r and c mean *row* and *column*; they do not denote a radius. The two matrices are

$$\mathbf{D}_r = \text{diag}(d_0^{\text{row}}, \dots, d_{26}^{\text{row}}), \quad \mathbf{D}_c = \text{diag}(d_0^{\text{col}}, \dots, d_{26}^{\text{col}}). \quad (44)$$

The source does not store 729 entries for either diagonal matrix. The arrays $\text{Dr}[27]$ and $\text{Dc}[27]$ store only d_i^{row} and d_i^{col} , respectively. They start at one. During each equilibration sweep, the code first divides every row by its largest absolute entry and accumulates that multiplier in Dr ; it then does the same for every column and accumulates the multiplier in Dc . In index notation, one sweep performs

$$\begin{aligned} s_i^{\text{row}} &= \max_j |\widetilde{M}_{ij}|, & \widetilde{M}_{i:} &\leftarrow \widetilde{M}_{i:}/s_i^{\text{row}}, & d_i^{\text{row}} &\leftarrow d_i^{\text{row}}/s_i^{\text{row}}, \\ s_j^{\text{col}} &= \max_i |\widetilde{M}_{ij}|, & \widetilde{M}_{:j} &\leftarrow \widetilde{M}_{:j}/s_j^{\text{col}}, & d_j^{\text{col}} &\leftarrow d_j^{\text{col}}/s_j^{\text{col}}. \end{aligned} \quad (45)$$

Thus $\mathbf{D}_r$ rescales the 27 equations (rows), whereas $\mathbf{D}_c$ rescales the 27 unknowns (columns). Both contain positive finite diagonal entries and are therefore nonsingular, so the exact rank is unchanged.

The rank gate uses only $\widetilde{\mathbf{M}}$. When a right-hand side is later solved, the same scaling is used consistently:

$$\widetilde{\mathbf{C}} = \mathbf{D}_r \mathbf{C}, \quad \widetilde{\mathbf{M}} \mathbf{y} = \widetilde{\mathbf{C}}, \quad \mathbf{\Psi} = \mathbf{D}_c \mathbf{y}. \quad (46)$$

Substitution gives $\mathbf{D}_r \mathbf{M} \mathbf{D}_c \mathbf{y} = \mathbf{D}_r \mathbf{C}$, which is equivalent to $\mathbf{M} \mathbf{\Psi} = \mathbf{C}$. Therefore equilibration changes only numerical scaling, not the original solution. The reason for testing $\widetilde{\mathbf{M}}$ is that entries associated with polynomial degrees one through six can have very different magnitudes; a QR test on the raw matrix can mistake this scale imbalance for rank loss.

The code then applies deterministic rank-revealing QR to $\widetilde{\mathbf{M}}^\top$. If p_i is the residual norm of the i th QR pivot, it is counted as an independent direction only when

$$p_i > \tau_{\widetilde{\mathbf{M}}}, \quad \tau_{\widetilde{\mathbf{M}}} = 27 \epsilon_{\text{dbl}} \|\widetilde{\mathbf{M}}\|_F. \quad (47)$$

The result is therefore interpreted as follows:

- if QR finds 27 such pivots, $\text{rank}_{\text{num}}(\widetilde{\mathbf{M}}) = 27$ and the moment-matrix rank gate passes;
- if QR finds fewer than 27, the current (stencil, β , h) trial is rejected and the selector proceeds to the next trial. No weights are exported.

Before this decisive test, the same QR routine also checks column-normalised versions of $\mathbf{X}$ and $\mathbf{W}$. These two preliminary tests identify whether rank loss already originates in polynomial sampling or in the Hermite–Wendland rows. Consequently every trial must pass

$$\text{rank}_{\text{num}}(\mathbf{X}) = 27, \quad \text{rank}_{\text{num}}(\mathbf{W}) = 27, \quad \text{rank}_{\text{num}}(\widetilde{\mathbf{M}}) = 27. \quad (48)$$

The first two conditions are necessary because $\text{rank}(\mathbf{X}^\top \mathbf{W}) \leq \min\{\text{rank}(\mathbf{X}), \text{rank}(\mathbf{W})\}$, but they do not guarantee the third. Hence the QR result for $\widetilde{\mathbf{M}}$ is the decisive rank test. This is a double-precision numerical certificate, not a symbolic proof of $\det(\mathbf{M}) \neq 0$.

Step 5: reciprocal-conditioning gate. The equilibrated matrix must satisfy

$$\text{rcond}_\infty(\widetilde{\mathbf{M}}) > \gamma_{27}, \quad \gamma_{27} = \frac{27\epsilon_{\text{dbl}}}{1 - 27\epsilon_{\text{dbl}}} \simeq 5.9952 \times 10^{-15}, \quad (49)$$

where ϵ_{dbl} is double-precision machine epsilon. This is a roundoff-level rejection threshold; it is not a claim that every accepted matrix has a small condition number. Here $\text{rcond}_\infty(\mathbf{B}) = 1/(\|\mathbf{B}\|_\infty \|\mathbf{B}^{-1}\|_\infty)$ for a nonsingular $\mathbf{B}$. The raw $\text{rcond}_\infty(\mathbf{M})$ is retained for diagnosis but is not an acceptance gate.

Step 6: direction-independent amplification gate. Define two 27-component probe right-hand sides. The Y probe $\mathbf{e}^{[Y]}$ is one at basis label $(1, 0)$ and zero elsewhere; the Z probe $\mathbf{e}^{[Z]}$ is one at $(0, 1)$ and zero elsewhere. For $\ell \in \{Y, Z\}$, solve

$$\mathbf{M}\Psi^{[\ell]} = \mathbf{e}^{[\ell]}, \quad w_n^{[\ell]} = \sum_{\alpha=1}^{27} W_{n\alpha} \Psi_\alpha^{[\ell]}, \quad w_0^{[\ell]} = 1 - \sum_{n=1}^{30} w_n^{[\ell]}. \quad (50)$$

The direction-independent amplification is

$$\lambda_0 = \max_{\ell \in \{Y, Z\}} \sum_{n=0}^{30} |w_n^{[\ell]}|. \quad (51)$$

The compact tier requires $\lambda_0 \leq 512$; repair trials require $\lambda_0 \leq 4096$. These limits reject extreme cancellation. They are ceilings, not quantities minimised by the selector.

Passing (48)–(51) is an a posteriori finite-precision certificate for the particular coordinates, h and anisotropic scales that were tested. It is not an analytic proof that every stencil proposed by the index-space pyramid is nonsingular. A failed compact trial is sent to the next (β, h) pair or to the repair tier; if every trial fails, no stencil is exported.

3.7.4 Repair tier and the leading-error objective

Purpose of this stage. This stage is a fallback, not a second optimisation pass over an already accepted template. It constructs alternative 30-neighbour clouds only after all compact trials fail, applies exactly the same solvability gates, and then ranks only the survivors by their leading degree-seven moment leakage.

When repair is entered. Repair is not run after a compact stencil has already passed. It is entered only when every combination

$$\beta \in \{0, \tfrac{1}{2}, 1\}, \quad h/h_{\text{templ}} \in \{1, 1.2, 1.5, 2\}$$

for the fixed pyramid has failed at least one admissibility gate in §3.7.3.

How the three repair sources construct 30 neighbours. Here QRCP means the deterministic rank-revealing QR procedure that treats physical points as row candidates and orders the selected rows by their independent contribution to the Taylor matrix. Let N_{pool} denote the number of retained pool points. In the table, a prefix of nominal length L means the first $\min(L, N_{\text{pool}})$ points.

Source	Construction	Points guaranteed in that trial
A: unchanged pyramid	Use exactly the same 30 neighbours as the compact template, but apply the repair-tier limit $\lambda_0 \leq 4096$.	All 30 template neighbours.
B: minimally modified pyramid	Run QRCP on the 30 template rows. If its returned rank r_{templ} is at least 24, retain the first $\min(r_{\text{templ}}, 27)$ template anchors. Exclude every other template point from refill, then add eligible non-template candidates in pool order until 30 points are present.	Between 24 and 27 template anchors; consequently 6 to 3 points are replaced.
C: pool-prefix cloud	Open the first 40, 60, 100 and 160 pool entries in turn. Among entries satisfying $q_n < 2$, QRCP must find exactly 27 independent anchors; the first three eligible non-anchors in pool order complete the cloud.	Exactly 27 pool-QRCP anchors and three deterministic local fill points.

Thus source B can replace three to six points. The nearby source-code comment that says at most three replacements is stale relative to the executable condition $r_{\text{templ}} \geq 24$.

Scale parameters tried by repair. Sources A and B reuse h_{templ} from (30) and try

$$h/h_{\text{templ}} \in \{1, 1.2, 1.5, 2\}.$$

For source C, let h_{base} be the RMS value in (30) computed instead from the first 30 candidate-pool entries at the current β . Source C tries

$$h/h_{\text{base}} \in \{1.2, 1.5, 2, 2.5, 3, 4, 5, 6\}.$$

All three sources are generated for each β in (28), and every complete trial must independently pass the support, rank, reciprocal-conditioning and λ_0 gates of §3.7.3.

How the winning repair cloud is chosen. The first-moment probe weights $w_n^{[Y]}$ and $w_n^{[Z]}$ were defined in (50). Using the scaled physical displacements $\bar{Y}_n, \bar{Z}_n$ from (29), define

$$T_7 = \sum_{\ell \in \{Y, Z\}} \sum_{\substack{a+b=7 \\ a, b \in \mathbb{N}_0}} \frac{1}{a! b!} \left| \sum_{n=1}^{30} w_n^{[\ell]} \bar{Y}_n^a \bar{Z}_n^b \right|. \quad (52)$$

The P_6 moment conditions control degrees zero through six, so degree seven is the first uncontrolled Taylor degree. The code evaluates (52) for every admissible source-A, source-B and source-C trial and retains the strictly smallest value. Because the comparison is strict, an exact tie retains the trial encountered first in the fixed loop order.

T_7 is a direction-independent comparison proxy based on the two first-moment responses. It is not the complete truncation error for a particular LBM departure point, and it is meaningful here only for comparing trials at the same arrival node. The older helper proportional to $\lambda_0 r_{\text{max}}^7$ has no caller in the active selector and is not the current optimisation objective.

3.7.5 Final direction-dependent solve and fail-closed export

Purpose of this stage. The preceding stages choose only the geometry. This final stage supplies one departure point at a time, computes that direction's weights on the already selected footprint, and prevents any failed solution from reaching the time-step loop.

One footprint, different weights for different lattice directions. Stages 1–4 select geometry without using an LBM velocity. Consequently one arrival node has one set of 31 storage IDs shared by all lattice directions. Direction-specific data enter only now through the departure point.

Let d denote an LBM direction and let $(e_y^{(d)}, e_z^{(d)})$ be its two lattice-velocity components. The physical displacement from the arrival node to the departure point is, with Δt denoting the global time step,

$$\delta_y^{(d)} = -\Delta t e_y^{(d)}, \quad \delta_z^{(d)} = -\Delta t e_z^{(d)}. \quad (53)$$

Use the same a_y, a_z, h as the accepted stencil and define

$$\xi_\delta^{(d)} = \frac{\delta_y^{(d)}}{a_y h}, \quad \eta_\delta^{(d)} = \frac{\delta_z^{(d)}}{a_z h}. \quad (54)$$

For the Taylor label $\mu = \mu(a, b)$ introduced in (33), the right-hand side is

$$C_{\mu(a,b)}^{(d)} = \frac{(\xi_\delta^{(d)})^a (\eta_\delta^{(d)})^b}{a! b!}. \quad (55)$$

This is the precise meaning of the abbreviated statement $\mathbf{C}^{(d)} = \mathbf{X}(\boldsymbol{\delta}^{(d)})$.

Solve for coefficients, then recover interpolation weights. For direction d , solve

$$\mathbf{M}\boldsymbol{\Psi}^{(d)} = \mathbf{C}^{(d)}. \quad (56)$$

The 30 neighbour weights and the structural centre weight are then

$$w_n^{(d)} = \sum_{\alpha=1}^{27} W_{n\alpha} \Psi_\alpha^{(d)} \quad (n = 1, \dots, 30), \quad w_0^{(d)} = 1 - \sum_{n=1}^{30} w_n^{(d)}. \quad (57)$$

Thus the footprint and $\mathbf{M}$ are shared across directions, whereas $\mathbf{C}^{(d)}$, $\boldsymbol{\Psi}^{(d)}$ and $\mathbf{w}^{(d)}$ depend on the departure direction.

Hard solve gates. The code equilibrates (56) and solves it independently by GEPP, explicit PLU and Gauss–Jordan. All three solutions undergo double-double iterative refinement using the factored residual of the original product $\mathbf{X}^\top \mathbf{W}$. Export is permitted only if

- (i) all three solvers report success;
- (ii) each solver’s normwise and componentwise backward errors are at most $64\gamma_{27}$;
- (iii) $\text{rcond}_\infty(\widetilde{\mathbf{M}}) > \gamma_{27}$; and
- (iv) for every component of $\boldsymbol{\Psi}$, each solver pair (s, t) satisfies $|\Psi_i^{(s)} - \Psi_i^{(t)}| / \max(1, |\Psi_i^{(s)}|, |\Psi_i^{(t)}|) \leq 64\gamma_{27}$.

In item (iv), $i = 1, \dots, 27$, and (s, t) runs over the three pairs GEPP–PLU, GEPP–Gauss–Jordan and PLU–Gauss–Jordan. Only after these tests does the code form (57) and allow `labfm_compact_export` to copy the 31 IDs and weights into the time-step data.

Warnings versus hard failures. After a successful solve, let $\mathbf{w}_{\text{nb}}^{(d)} = (w_1^{(d)}, \dots, w_{30}^{(d)})^\top$ denote the neighbour weights without the centre weight. The code also reports

$$R_d = \frac{\|\mathbf{X}^\top \mathbf{w}_{\text{nb}}^{(d)} - \mathbf{C}^{(d)}\|_2}{\|\mathbf{C}^{(d)}\|_2}, \quad \Lambda_d = \sum_{n=0}^{30} |w_n^{(d)}|, \quad (58)$$

with the zero denominator treated as one for the rest direction. The thresholds $R_d > 10^{-8}$ and $\Lambda_d > 6$ currently set warning bits only; they are not the same as the selection hard gates (49) and (51).

If point selection or any hard solve gate fails, no compact stencil is exported. The active path has no pseudoinverse, regularisation, wall-ghost substitution or nearest-neighbour fallback. A concise Chinese source map is maintained separately in `labfm/stencil.md`.

3.8 A parity diagnostic for sampling adequacy

The angular sampling condition of §3.6 is stated as a geometric rule and measured in [1] through the condition number, which is a lagging indicator: by the time κ jumps, undersampling is already severe. A direct and inexpensive leading indicator follows from the parity structure of the continuum limit [N].

For the Hermite construction, He_n has parity $(-1)^n$, so the integrand of $\langle X^\alpha, W^\beta \rangle$ with $\alpha = (a, b, c)$, $\beta = (a', b', c')$ is odd in at least one coordinate unless every one of $a+a'$, $b+b'$, $c+c'$ is even. On an isotropic, symmetric support the integral then vanishes:

$$\langle X^\alpha, W^\beta \rangle = 0 \quad \text{unless } a+a', b+b', c+c' \text{ are all even.} \quad (59)$$

This is the analytic content of the pattern of machine-zero entries displayed in [1, Fig. 5]. Define the *parity defect*

$$\eta_i = \left(\frac{\sum_{(\alpha, \beta) \text{ forbidden}} |M_i^{\alpha\beta}|^2}{\sum_{\alpha, \beta} |M_i^{\alpha\beta}|^2} \right)^{1/2}, \quad (60)$$

computable in $O(p^2)$ with no quadrature. It serves three purposes: on a uniform Cartesian cloud $\eta_i \lesssim 10^{-14}$ is a sharp test of the assembly (a failure indicates a multi-index ordering or Hermite recurrence error); the value of h/s at which η_i begins to rise locates the critical ratio ahead of any change in κ ; and a large η_i at a node correctly flags an anisotropic or one-sided stencil, identifying where the order must be reduced.

Where an absolute rather than relative measure is wanted, the continuum reference $N_i \langle X^\alpha, W^\beta \rangle$ of (14) may be evaluated by quadrature over the support and compared with the assembled M_i .

3.9 Published stability results

For completeness, and because they bound what may legitimately be assumed:

- On ideal Cartesian distributions the convective operator has $\Re(\lambda) \approx 0$ provided $dW^0/dr = 0$ on the support boundary, a condition satisfied by the quadratic and Wendland fundamental functions but not by the conic one [1, §4.2]. On distributions carrying disorder, $\Re(\lambda) > 0$ for all choices tested.
- The Laplacian operator is stable, but the admissible level of distribution noise decreases as m increases [1, §4.2, Table 1].
- Hyperviscosity operators up to ∇^m may be generated from the same system (6) by an appropriate choice of C^{hv} [1, Eq. (25)].

Remark 8. *These results concern differential operators and therefore do not transfer directly to the present scheme, in which streaming is effected by interpolation. The governing stability condition for the scheme specified here is (68) below.*

3.10 Boundaries: what the literature actually reports

This subsection supersedes the abstract of [1] and is the single most consequential correction in this revision.

The 2020 claim. [1] states in its abstract that “at domain boundaries (with incomplete support) LABFM automatically provides one-sided differences of the same order as the internal scheme, up to 4th order”, and in [1, §IV A] that convergence at order m is retained near boundaries provided the distribution noise is small enough and h/s large enough.

The 2021 retraction. [2, §7.1], by the same authors, reports that this does not hold:

- for one-sided stencils and $m > 4$, “the conditioning of the linear systems M_i was such that the solver didn’t yield a valid solution”;
- “a new stability analysis since [1] (via the eigenvalues of the global derivative operator) has shown that for one-sided LABFM with $m = 4$ the eigenvalues spread along the positive real axis: *the operators are unstable*”;
- for nearly one-sided stencils one node in from the boundary, the viscous operators are stable but the convective operators remain unstable.

[2] accordingly *abandons* one-sided LABFM and adopts a hybrid, which requires a strip of five uniformly distributed, locally orthogonal nodes along every boundary:

Node	$\partial/\partial\xi$ (normal)	$\partial/\partial\eta$ (tangential)	Second derivatives
i_0 (on boundary)	one-sided FD	1D-LABFM	FD + 1D-LABFM
i_1	asymmetric FD	1D-LABFM	LABFM, $m = 4$
i_2	centred FD	1D-LABFM	LABFM, $m = 4$
i_3, i_4 , interior	LABFM	LABFM	LABFM

([2, Table 1]; the one-dimensional formulation used for tangential derivatives is given in [2, Appendix A, Eqs. (A1)–(A7)], and first derivatives at i_1, i_2 are rotated back to Cartesian components through the Jacobian [2, Eq. (24)].) The production configuration of [2] is therefore $m = 6$ in the interior and an effectively fourth-order treatment in the boundary strip.

Consequences for the present scheme. Four, and they must be kept distinct.

- (i) **The conditioning risk transfers, but the categorical stencil conclusion does not.** For a fixed cloud, basis and scaling, M_i is independent of the right-hand side, so a singular derivative construction cannot be rescued merely by asking it to interpolate. This does not prove that every other degree-six physical cloud is singular. The current selector of §3.7 changes the cloud, permits a controlled anisotropic coordinate scaling, and requires the actual one-sided $\mathbf{X}$, $\mathbf{W}$ and equilibrated $\mathbf{M}$ to pass rank and reciprocal-condition gates. It therefore treats the 2021 result as a fail-closed warning to test each wall cloud, not as a theorem that $m > 4$ is impossible for all interpolation selectors.
- (ii) **The instability result does not transfer directly.** It is an eigenvalue statement about the global *derivative* matrix. The present scheme streams by interpolation, whose stability is governed by (68). Whether one-sided interpolation operators are admissible must therefore be settled by measuring $\max_{\mathbf{k}} |G_q(\mathbf{k})|$ on one-sided stencils, not inferred from [2].
- (iii) **The current LABFM footprint is physical and one-sided at a wall.** Wall-normal ghosts are forbidden by (20); the seven-layer template is clamped into the domain. Ghost layers remain necessary for the incumbent ITB/GILBM paths, boundary closure and statistics contract, but they are not LABFM stencil members. If the finite-difference hybrid of [2] is later adopted, it is a different method branch rather than the selector presently encoded in `Lagrange_vs_LABFM/2.labfm/`.
- (iv) **Local algebraic acceptance is not global stability or physical validation.** Passing (48)–(51) and the three-solver gate establishes that the local interpolation weights are numerically constructible. It does not establish the one-sided global amplification bound, observed

seventh-order convergence at every wall geometry, or correctness of a wall-resolved production run.

Open item U4 records what must be measured before a wall-resolved production claim is made.

4 The departure-point interpolation operator

4.1 Definition

Because the characteristics of the free-streaming operator are straight lines in physical space, the departure point is available in closed form,

$$\delta_q = -\mathbf{e}_q c \Delta t, \quad \mathbf{r}_q^* = \mathbf{r}_i + \delta_q, \quad (61)$$

with c the lattice speed and Δt the global time step. No integration of the characteristic and no metric information is required.

Remark 9 (The three points, and why the departure point carries no index). *Three locations appear and are easily conflated. The node $\mathbf{r}_i$ is the Taylor origin, the stencil centre, and the arrival node at which the result is stored; its role is exactly that of i in [1] and is unchanged. The nodes $\mathbf{r}_j$, $j \in \mathcal{N}_i$, supply data; they are neighbours, and none of them is the departure point. The departure point $\mathbf{r}_q^*$ is neither: it is a query location that enters only through the right-hand side of (63). It does not appear in $\mathbf{M}_i$ at all, which is precisely why one factorisation serves all nineteen discrete velocities (Remark 1).*

Proposition 3 (Admissibility of the node-centred stencil; [N]). *Centring the stencil at $\mathbf{r}_i$ requires the departure point to lie within the support, $|\delta_q| \leq 2h_i$. With $\Delta t = \text{CFL} \cdot s_{\min}/c$ and $\max_q |\mathbf{e}_q| = \sqrt{2}$ for D3Q19, and taking the worst case $s_i = s_{\min}$,*

$$\text{CFL} \leq \sqrt{2} (h/s). \quad (62)$$

At $h/s = 1.6$ this gives $\text{CFL} \leq 2.26$; a working margin of one half of the support radius halves it. The present operating point, $\text{CFL} = 0.5$, places the departure point at between 2 % and 22 % of the support radius depending on local spacing, comfortably inside. The bound is recorded because it is not independent of the time-step question: the relaxation-time constraint $\tau = 3\nu/\Delta t + \frac{1}{2}$ becomes binding at a comparable value, so any attempt to enlarge Δt will approach both limits at once. Beyond (62) the stencil would have to be centred on the departure point, at the cost of one moment matrix per discrete velocity (Remark 4).

Definition 1 (Departure-point interpolation operator; [N]). *Using the same moment matrix $\mathbf{M}_i$ as in (6), define for each discrete velocity q*

$$\mathbf{M}_i \Psi_i^q = \widetilde{\mathbf{X}}(\delta_q) \equiv \mathbf{X}(\delta_q) \cdot \mathbf{H}_i, \quad w_{ji}^q = \mathbf{W}_{ji} \cdot \Psi_i^q, \quad (63)$$

and take as the streaming step

$$f_q(\mathbf{r}_i, t + \Delta t) = f_{q,i}^{\text{post}} + \sum_{j \in \mathcal{N}_i} (f_{q,j}^{\text{post}} - f_{q,i}^{\text{post}}) w_{ji}^q = w_{\text{cen}}^q f_{q,i}^{\text{post}} + \sum_{j \in \mathcal{N}_i} w_{ji}^q f_{q,j}^{\text{post}}, \quad (64)$$

where $w_{\text{cen}}^q \equiv 1 - \sum_j w_{ji}^q$.

The form of the right-hand side follows from applying (3) to the departure point, $\phi(\mathbf{r}_q^*) - \phi_i = \mathbf{X}(\delta_q) \cdot \mathbf{D}\phi|_i$, and comparing with (4). Consistency of the preconditioning is guaranteed by using the same $\mathbf{H}_i$ in $\widetilde{\mathbf{X}}(\delta_q)$ as in $\widetilde{\mathbf{X}}_{ji}$.

4.2 Polynomial reproduction

Theorem 1 (Exact reproduction at the departure point; [N]). *Let $\mathbf{M}_i$ be non-singular, let $\mathbf{X}$ contain all monomials of total degree at most m , and let Ψ_i^q solve (63). Then for every polynomial P of total degree at most m ,*

$$P_i + \sum_{j \in \mathcal{N}_i} (P_j - P_i) w_{ji}^q = P(\mathbf{r}_i + \delta_q),$$

exactly, with no truncation error.

Proof. For a polynomial of total degree at most m the truncated expansion (3) is an identity rather than an approximation, so that $P_j - P_i = \mathbf{X}_{ji} \cdot \mathbf{D}P|_i$. Hence

$$\begin{aligned} \sum_j (P_j - P_i) w_{ji}^q &= \sum_j (\mathbf{X}_{ji} \cdot \mathbf{D}P|_i) (\mathbf{W}_{ji} \cdot \Psi_i^q) = \mathbf{D}P|_i \cdot \left(\sum_j \mathbf{X}_{ji} \otimes \mathbf{W}_{ji} \right) \Psi_i^q \\ &= \mathbf{D}P|_i \cdot \mathbf{M}_i \Psi_i^q = \mathbf{D}P|_i \cdot \mathbf{X}(\delta_q) = P(\mathbf{r}_i + \delta_q) - P(\mathbf{r}_i), \end{aligned}$$

the last step again by the exactness of (3) for polynomials of degree at most m . Adding P_i gives the result. The preconditioned form is equivalent, since both sides of the linear system are multiplied by the same diagonal matrix. $\square$

Corollary 2 (Exact preservation of constants). *Taking $P \equiv \text{const}$ gives $P_j - P_i \equiv 0$, so that (64) returns $f_{q,i}$ identically. A uniform field therefore generates no spurious mass source. This follows from the difference form $(\cdot)_{ji}$ alone and requires no additional partition-of-unity constraint on the weights.*

Corollary 3 (Degenerate zero-displacement limit). *If $\delta_q = \mathbf{0}$ then $\mathbf{X}(\mathbf{0}) = \mathbf{0}$, hence $\Psi_i^q = \mathbf{0}$ and $w_{ji}^q \equiv 0$, and (64) reduces to the identity exactly. This provides the sharpest available unit test of the assembled operator (rung L5a of §6).*

Remark 10 (The operator is approximating, not interpolating). *Lagrange interpolation reproduces nodal values exactly: a departure point coinciding with a node returns that node's value. The operator of Definition 1 does not have this property, since it enforces polynomial consistency rather than nodal collocation, in common with moving least squares approximation. Consequently the condition “the scheme must recover the standard lattice Boltzmann method when the departure point coincides with a node” is not a valid verification criterion for this scheme. The only exact degeneracy is Corollary 3.*

4.3 Truncation error and the amplification constant

For a general smooth field, $f_j - f_i = \mathbf{X}_{ji} \cdot \mathbf{D}f|_i + R_{ji}$ with $R_{ji} = O(|\mathbf{r}_{ji}|^{m+1})$. Theorem 1 then gives

$$\varepsilon_i^q = \left| \sum_j R_{ji} w_{ji}^q - R(\delta_q) \right| \leq \Lambda_i^q \max_j |R_{ji}| + |R(\delta_q)| = O(\Lambda_i^q h_i^{m+1}), \quad (65)$$

where

$$\Lambda_i^q \equiv \sum_{j \in \mathcal{N}_i} |w_{ji}^q| \quad (66)$$

plays the role of a Lebesgue constant for the operator [N]. It bounds both the error amplification in (65) and the growth of the streaming step in the maximum norm, since (64) gives $|f^{\text{new}}| \leq (|w_{\text{cen}}^q| + \Lambda_i^q) \max_j |f_j|$. The quantity Λ_i^q is to be computed and reported for every node and every discrete velocity during preprocessing; it is the second diagnostic, alongside the condition number, on which acceptance of a stencil is conditioned.

4.4 Von Neumann amplification and the stability criterion

Substituting a plane wave $f = \exp(i\mathbf{k} \cdot \mathbf{r})$ into (64) gives the amplification factor

$$G_q(\mathbf{k}) = 1 + \sum_{j \in \mathcal{N}_i} \left(e^{i\mathbf{k} \cdot \mathbf{r}_{ji}} - 1 \right) w_{ji}^q, \quad (67)$$

whose exact counterpart is the pure phase shift $e^{i\mathbf{k} \cdot \boldsymbol{\delta}_q}$, of unit modulus. Hence:

[N] Stability criterion for the streaming step.

$$\max_{|\mathbf{k}| \leq k_{\text{Ny}}} |G_q(\mathbf{k})| \leq 1 + \epsilon_{\text{tol}}, \quad q = 0, \dots, 18, \quad k_{\text{Ny}} = \pi/s_i. \quad (68)$$

The criterion is evaluated per node during preprocessing. A node violating it must have its polynomial order reduced or its stencil enlarged. The phase error $\arg G_q(\mathbf{k}) - \mathbf{k} \cdot \boldsymbol{\delta}_q$ is recorded separately as the dispersion diagnostic.

Remark 11 (Relation to the source). *Criterion (68) is an adaptation of steps 3 and 4 of the stencil optimisation algorithm of [2, §4], in which the amplitude response of the differential operators is evaluated at $n_k = 16$ wavenumbers up to the Nyquist limit [2, Eqs. (13a)–(13c)] and the stencil is accepted only while $\max(A^x, A^y, A^L) \leq 1.01$. The present criterion replaces the effective wavenumber of a derivative by the amplification factor of a translation; the tolerance $\epsilon_{\text{tol}} = 0.01$ and the sampling $n_k = 16$ are retained from the source. The same Fourier machinery underlies the resolving-power analysis of [3, 4], which provides the natural route to a future implicit (compact) variant.*

4.5 Local affine rescaling for anisotropic node distributions

Near a wall the node spacing normal to the surface is much smaller than the tangential spacing, so the stencil is a flattened ellipsoid in physical space and $\mathbf{M}_i$ is poorly conditioned. Let $\mathbf{S}_i = \text{diag}(h_{x,i}^{-1}, h_{y,i}^{-1}, h_{z,i}^{-1})$ and assemble and solve the system in the rescaled coordinates $\tilde{\mathbf{r}}_{ji} = \mathbf{S}_i \mathbf{r}_{ji}$.

Theorem 4 (Affine invariance of polynomial consistency; [N]). *The space $\mathbb{P}_m$ of polynomials of total degree at most m is invariant under any invertible linear map. Consequently, enforcing degree- m consistency in $\tilde{\mathbf{r}}$ is equivalent to enforcing degree- m consistency in $\mathbf{r}$.*

Proof. Let A be invertible and $P \in \mathbb{P}_m$. Then $P \circ A$ is a polynomial with $\deg(P \circ A) = \deg P$, since each monomial of total degree n maps to a linear combination of monomials of total degree n . Theorem 1 applied to the point set $\{\mathbf{A}\mathbf{r}_{ji}\}$ with right-hand side $\mathbf{X}(\mathbf{A}\boldsymbol{\delta}_q)$ is therefore equivalent to the same statement for $\{\mathbf{r}_{ji}\}$ and $\mathbf{X}(\boldsymbol{\delta}_q)$. $\square$

Remark 12 (Implementation consequences). *Interpolation weights are dimensionless, so no inverse scaling of the weights is required; it suffices to evaluate the right-hand side as $\widetilde{\mathbf{X}}(\mathbf{S}_i \boldsymbol{\delta}_q)$. Weights of differential operators do carry dimensions and must be rescaled by the corresponding powers of $\mathbf{S}_i$. The Hermite ABFs (12) extend to three dimensions as*

$$W_{ji}^\alpha = \frac{\psi(|\tilde{\mathbf{r}}_{ji}|)}{\sqrt{2^{a+b+c}}} H_a\left(\frac{\tilde{x}_{ji}}{\sqrt{2}}\right) H_b\left(\frac{\tilde{y}_{ji}}{\sqrt{2}}\right) H_c\left(\frac{\tilde{z}_{ji}}{\sqrt{2}}\right), \quad (69)$$

for $\mathbf{X}$ -entries with multi-index $\alpha = (a, b, c)$; this is the direct dimensional extension of [2, Eq. (10)] [N]. Throughout, a Greek superscript indexes the basis, running over $1, \dots, p$, while the Latin q reserved in §4 indexes the discrete velocity; the two are unrelated.

5 A generalised moving least squares control formulation

The construction of §4 uses the LABFM machinery, but nothing in Theorems 1 and 4, in Corollaries 2 and 3, or in the diagnostics (66) and (68) requires it: they apply to any weight construction that enforces polynomial consistency on a node cloud. The most established such alternative is moving least squares [6] in its generalised, node-centred form [7, 8]. This section specifies that alternative in a form that is directly comparable, term by term, so that the selection between the two can be made by measurement rather than by assertion.

5.1 Definition

Let $\mathbf{q}(\mathbf{z}) = [1, z_x, z_y, z_z, z_x^2/2, \dots]^\top$ be the complete polynomial basis of total degree at most m , of dimension $P = \binom{m+3}{3} = p_{3D} + 1$, and let $\theta_{ji} = \theta(|\mathbf{r}_{ji}|/h_i)$ be a compactly supported kernel. Write $\mathbf{q}_{ji} = \mathbf{q}(\mathbf{S}_i \mathbf{r}_{ji})$ with $\mathbf{S}_i$ the local affine scaling of §4.5. Define the Gram matrix and the shape functions evaluated at the departure point,

$$\mathbf{A}_i = \sum_{j \in \mathcal{N}_i} \theta_{ji} \mathbf{q}_{ji} \mathbf{q}_{ji}^\top, \quad \Phi_{ji}^q = \mathbf{q}(\mathbf{S}_i \boldsymbol{\delta}_q)^\top \mathbf{A}_i^{-1} \theta_{ji} \mathbf{q}_{ji}. \quad (70)$$

Remark 13 (Why the kernel is centred on the node). *In classical moving least squares the kernel is centred on the evaluation point, so that $\mathbf{A}$ must be reassembled and refactorised for every target. With nineteen targets per node this would forfeit the structural advantage identified in Remark 1. Centring the kernel on node i instead — the generalised or fixed-kernel variant [7, 8] — makes $\mathbf{A}_i$ target-independent, so that a single factorisation serves all nineteen departure points, exactly as $\mathbf{M}_i$ does. Only this variant constitutes a fair control.*

Because $1 \in \text{span } \mathbf{q}$, the shape functions form a partition of unity, $\sum_j \Phi_{ji}^q = 1$, and the streaming step may be written in the same difference form as (64) with

$$w_{ji}^{q, \text{MLS}} = \Phi_{ji}^q. \quad (71)$$

The diagnostics (66) and (67) therefore apply verbatim to both arms, with identical definitions.

5.2 Identical formal order

Theorem 5 (Equivalence of polynomial reproduction; [N]). *Let $\mathbf{A}_i$ be non-singular. Then for every polynomial P of total degree at most m ,*

$$\sum_{j \in \mathcal{N}_i} \Phi_{ji}^q P(\mathbf{r}_j) = P(\mathbf{r}_i + \boldsymbol{\delta}_q),$$

exactly. Consequently the operator (71) has the same leading truncation error $O(h_i^{m+1})$ as the operator of Definition 1.

Proof. Since $\mathbf{q}$ spans $\mathbb{P}_m$ in the scaled variable, there exists $\mathbf{c}$ with $P(\mathbf{r}_j) = \mathbf{q}_{ji}^\top \mathbf{c}$ for all j , and $P(\mathbf{r}_i + \boldsymbol{\delta}_q) = \mathbf{q}(\mathbf{S}_i \boldsymbol{\delta}_q)^\top \mathbf{c}$. Then

$$\sum_j \Phi_{ji}^q P(\mathbf{r}_j) = \mathbf{q}(\mathbf{S}_i \boldsymbol{\delta}_q)^\top \mathbf{A}_i^{-1} \sum_j \theta_{ji} \mathbf{q}_{ji} \mathbf{q}_{ji}^\top \mathbf{c} = \mathbf{q}(\mathbf{S}_i \boldsymbol{\delta}_q)^\top \mathbf{A}_i^{-1} \mathbf{A}_i \mathbf{c} = \mathbf{q}(\mathbf{S}_i \boldsymbol{\delta}_q)^\top \mathbf{c}. \quad \square$$

Consequence for the contribution statement. Theorem 5 shows that polynomial consistency of degree m , and hence interpolation of order $m + 1$, is a property of the consistency constraint and not of the LABFM basis. Any claim that high-order consistency on a node cloud is specific to LABFM is therefore unfounded and must not be made. What existing meshless lattice Boltzmann formulations lack is not the capability but the realisation: the highest polynomial augmentation reported in [12] is degree four, in two dimensions.

5.3 The single structural difference: conditioning

Stack the rows $\mathbf{q}_{ji}^\top$ into $\mathbf{Q} \in \mathbb{R}^{N_i \times P}$ and let $\boldsymbol{\Theta} = \text{diag}(\theta_{ji})$. Then

$$\mathbf{A}_i = \mathbf{Q}^\top \boldsymbol{\Theta} \mathbf{Q} = (\boldsymbol{\Theta}^{1/2} \mathbf{Q})^\top (\boldsymbol{\Theta}^{1/2} \mathbf{Q}), \quad \text{whence} \quad \kappa_2(\mathbf{A}_i) = \kappa_2(\boldsymbol{\Theta}^{1/2} \mathbf{Q})^2. \quad (72)$$

The moving least squares system is a Gram matrix, formed by normal equations, and its condition number is therefore the *square* of that of the underlying weighted design matrix. By contrast, stacking $\widehat{\mathbf{X}}_{ji}^\top$ into $\mathbf{X} \in \mathbb{R}^{N_i \times p}$ and $\mathbf{W}_{ji}^\top$ into $\mathbf{W}$,

$$\mathbf{M}_i = \mathbf{X}^\top \mathbf{W}, \quad (73)$$

a product of two *distinct* matrices — a Petrov–Galerkin rather than a Galerkin construction — which is not a Gram matrix and involves no squaring. In the language of §3.3 the two arms are the two selectors of (10), acting on one and the same constraint set (9); by Proposition 1 they even coincide when $N_i = p$, so the entire difference between them lies in how the surplus $N_i - p$ degrees of freedom are spent. The Hermite ABFs (12) are moreover orthogonal with respect to a Gaussian weight while $\mathbf{X}$ consists of monomials, so that $\mathbf{M}_i$ is close to block diagonal for near-uniform clouds; the corresponding structure, including entries that vanish to machine precision by parity, is displayed in [1, Fig. 5].

[N] Hypothesis H1 (to be tested at G1). On the production node cloud and at matched m , h/s , kernel and scaling, $\kappa(\mathbf{A}_i) \sim \kappa(\mathbf{M}_i)^2$, with the gap widening as m increases; and the largest admissible m is correspondingly larger for the LABFM arm.

H1 is the only structural discriminator between the two arms that survives Theorem 5. If it is not supported by measurement, there is no theoretical ground for preferring LABFM over moving least squares in this application.

5.4 Properties that are not discriminators

The following are shared by both arms and must not be advanced as reasons to prefer either:

- (a) formal order of accuracy (Theorem 5);
- (b) a single factorisation per node serving all nineteen departure points (Remark 1 for LABFM; the node-centred kernel for MLS);
- (c) automatic one-sided operators where the support is incomplete, provided the local matrix remains non-singular;
- (d) exact preservation of constants (Corollary 2 by the difference form for LABFM; the partition of unity for MLS);
- (e) applicability of the diagnostics Λ_i^q and $\max_{\mathbf{k}} |G_q(\mathbf{k})|$.

Preprocessing cost in fact favours the control arm: $\mathbf{A}_i$ is symmetric positive definite and admits a Cholesky factorisation at $O(P^3/3)$, whereas $\mathbf{M}_i$ is non-symmetric and requires $O(2p^3/3)$.

5.5 Fairness conditions

The comparison is invalid unless all of the following hold. They are stated here so that a failure to satisfy them is detectable in review.

- (F1) Identical node cloud, identical stencil membership, identical support radius $2h_i$ and identical h/s .
- (F2) Both arms use the local affine rescaling $\mathbf{S}_i$ of §4.5.
- (F3) Both arms use a shifted and scaled polynomial basis: the row preconditioning (13) for LABFM, and the scaled argument $\mathbf{q}(\mathbf{S}_i \mathbf{r}_{ji})$ for MLS. Comparing a preconditioned LABFM system against an unscaled MLS system is a straw man.

- (F4) Identical kernel, $\psi = \theta = \text{Wendland C2}$.
(F5) Identical diagnostics, computed by the same code path: κ , the residual of the local solve, Λ_i^q , $\max_{\mathbf{k}} |G_q(\mathbf{k})|$ at $n_k = 16$, and the monomial reproduction residual of Theorem 1 respectively Theorem 5.
(F6) Identical arithmetic precision and identical factorisation library.

5.6 Decision rule

At gate G1 both arms are run on the production node cloud and the following are reported for each: the largest m admitting $\kappa < 10^{10}$; $\kappa(m)$ at the target order; Λ_i^q and $\max_{\mathbf{k}} |G_q|$ distributions; the factorisation cost; and the stencil radius in index units. Then

- if the LABFM arm admits a larger m , or exhibits a condition-number advantage of at least two orders at the target m , adopt LABFM; H1 is supported;
- if the two arms are comparable, adopt the control arm — it is simpler, cheaper to factorise and more widely implemented — and drop the LABFM designation;
- if the control arm is superior, adopt it.

In all three outcomes the theory of §4 is unaffected: Theorems 1 and 4, Corollaries 2 and 3, the amplification constant (66) and the stability criterion (68) hold verbatim for either construction, and the contribution statement of §1.3 is method-agnostic.

6 Verification ladder

Each rung reproduces either a published result or an identity proved above. A rung must pass before the next is attempted.

Rung	Target	Pass condition
L0	[1, Eqs. (15),(16)]	Basis-dimension generator returns $\{5, 14, 27, 44\}$ in two dimensions for $m = 2, 4, 6, 8$ and $\{3, 9, 19, 34, 55\}$ in three dimensions for $m = 1, \dots, 5$.
L0b	Eq. (59)	On a uniform Cartesian cloud the parity defect (60) satisfies $\eta_i \lesssim 10^{-14}$. Cheapest available test of the multi-index ordering and the Hermite recurrence; requires no solve.
L1	[1, §4.1]	Two dimensions, original conic ABFs, distribution noise $\varepsilon/s = 0.5$, $h/s = 2$, test function [1, Eq. (23)]: gradient converges at order m and the Laplacian at order $m - 1$ for $m = 2, 4, 6$. The truncation of Remark 5 must be applied or the numbers will not match.
L2	[1, §3.2]	Two-dimensional critical ratios recovered with the <i>original</i> ABFs: $(h/s)_{\text{crit}} \approx \{0.75, 1.15, 1.60, 2.10\}$ and $N_{\text{crit}} \approx \{8, 21, 37, 57\}$. Establishes the measurement procedure later applied to the Hermite construction (open items U1, U8).
L3	[2, §3]	<i>Negative test.</i> Three dimensions, original ABFs, $m = 4$: $\mathbf{M}$ of size 34 with rank 33, independently of the node distribution. A full-rank result indicates an error in the ABF assembly, not an improvement.
L4	[2, §3, Fig. 2]	Hermite ABFs reduce $\kappa(\mathbf{M})$ by several orders of magnitude relative to the original ABFs, and restore full rank in three dimensions at $m = 4$.
L5a	Corollary 3	$\delta_q = \mathbf{0}$ yields $w^q \equiv 0$ and an unchanged field to machine precision.
L5b	Theorem 1	All monomials of total degree at most m reproduced at the departure point to a residual below 10^{-12} .
L5c	Corollary 2	A uniform field is unchanged after streaming in all nineteen directions.
L5d	Theorem 5	<i>Control arm.</i> The moving least squares operator (71) reproduces all monomials of total degree at most m at the departure point to a residual below 10^{-12} , and satisfies $\sum_j \Phi_{ji}^q = 1$ to machine precision.

continued on next page

Rung	Target	Pass condition (cont.)
L5e	Proposition 1	With $N_i = p$ the weights equal $\mathbf{X}^{-\top}\mathbf{C}$ and are independent of the selector: both arms, and any choice of ABF, must agree bitwise, and in one dimension must reproduce the Lagrange weights on $m + 1$ points. Validates the assembly without exercising the basis at all.
L5f	Proposition 2	<i>Gauge test.</i> Rebuilding $\mathbf{X}$ and $\mathbf{C}$ about a displaced origin $\mathbf{r}_0 = \mathbf{r}_i + \varepsilon\mathbf{n}$, $\varepsilon \sim h$, with $\mathbf{S}$ and N_i unchanged, must leave $\mathbf{w}$ invariant to a relative 10^{-10} . Isolates errors in the monomial assembly and multi-index ordering, which rung L5b detects only at the highest degrees.
L6	[1, §4.2, Table 1]	Trend of the Laplacian stability limit with h/s and m recovered; corroborates the ABF implementation.
L7	Eq. (68)	On the production node cloud, $\max_{\mathbf{k}} G_q(\mathbf{k}) \leq 1.01$ for all nodes and all q .
L7b	§§3.7, 3.10	<i>Boundary admissibility.</i> On every one-sided and nearly one-sided P6/31 cloud, report the three numerical ranks, equilibrated rcond, λ_0 , T_7 , all final-solve gates and the direction-specific Λ_q ; then measure $\max_{\mathbf{k}} G_q(\mathbf{k}) $ for the assembled interpolation operator. Local gates settle constructibility, while the amplification test settles whether the derivative-instability result transfers; together they close open item U4.
L8	Hypothesis H1, §5.6	<i>Head-to-head.</i> Both arms run on the production node cloud under the fairness conditions of §5.5; the decision rule of §5.6 is applied and its outcome recorded.

7 Open items

No item in this table may be used as an implementation assumption until it has been measured and the result written back into this document.

#	Item	Why open, and how it is to be closed
U1	$(h/s)_{\text{crit}}^{\text{3D}}$	Not in the literature; [2] defers three dimensions. Measured on the production grid for $m = 6$ with a metric-ball support: rank $\mathbf{X} = 75$ –80 of 83 at $h/s = 1.5$ and 83 at $h/s \geq 1.6$ ($N_i = 142$), so $(h/s)_{\text{crit}}^{\text{3D}} \simeq 1.6$ for a lattice cloud. Whether this constant enters a future full three-dimensional path depends on a selector that is not implemented. The current two-dimensional P6/31 selector is fully specified in §3.7 and does not assume this three-dimensional value. [N]
U2	Unisolvency of structured lattice balls	Closed, negatively. The 123-point radius-three lattice ball at $h/s = 1.5$ attains rank $\mathbf{X} = 77$ of 83; the singular spectrum has a hard gap ($\sigma_{79}/\sigma_1 = 4.9 \times 10^{-8}$, $\sigma_{80}/\sigma_1 = 1.0 \times 10^{-17}$), so the deficiency is exact and not a threshold artefact. It occurs at mid-channel and at the flat upper wall as well as on the hill face, hence it is a property of the lattice ball and not of the metric. A sufficient design screen is the slicing count: the support must decompose into $m + 1$ parallel slices holding at least $\dim \mathbb{P}_{m-\ell}^{\text{2D}} = 28, 21, 15, 10, 6, 3, 1$ points; this predicts the measured outcome at $h/s = 1.5, 1.6$ and 1.8 correctly. [N]
U3	Admissible bound on Λ_i^q	No counterpart exists in the literature, which does not define an interpolation operator. Reference values are now available on a 25-point in-plane support: 1.2–2.4 in the interior, 3.0 in the wall band; a rejected configuration reaches 3.5×10^4 and a ghost-fed one 1.2×10^3 . The current P6/31 implementation uses a distinct, direction-independent hard gate $\lambda_0 \leq 512$ in the compact tier and $\lambda_0 \leq 4096$ in repair, while the actual direction-specific $\Lambda_i^q > 6$ remains a warning rather than rejection. The admissible production bound on Λ_i^q is therefore still open and must be fixed by the error and stability budget. Close jointly with rung L7 and U5.

continued on next page

#	Item	Why open <i>(cont.)</i>
U4	Near-wall treatment	Partly closed at the local-algebra level. The current selector uses no wall ghost: it clamps a physical seven-layer P6/31 cloud into the domain and fails closed unless the rank, equilibrated-rcond, λ_0 and three-solver gates of §3.7 pass. This addresses numerical constructibility for the audited grids and avoids treating the adverse result of [2, §7.1] as a universal impossibility theorem. Still open are the global interpolation amplification spectrum, observed wall convergence over the supported grid family, and production physical validation. Close by rung L7b; until then, local gate success must not be reported as proof of wall-stable seventh-order production accuracy.
U5	Interpolation error against the lattice Boltzmann error floor	Determines whether raising the interpolation order changes the solution at all, given the saturation law of [12]. Close by an offline measurement on a converged field.
U6	Measured roofline position	Determines whether a reduction in operation count converts to wall-clock time. Close by profiling the existing kernel.
U7	Hypothesis H1: which weight construction	Theorem 5 removes order of accuracy as a discriminator, leaving conditioning (72) versus (73) as the only structural difference. Close by rung L8 under the fairness conditions of §5.5; the decision rule is §5.6. Until U7 is closed, the method designation in all external material is to be “consistency-based meshless departure-point interpolation”, not “LABFM”.
U8	Critical ratio for the Hermite ABFs	The published critical values belong to the <i>original</i> ABFs, which the production path does not use; only recommended operating values with an unquantified margin are published for the Hermite construction (§3.6). The present P6/31 implementation bypasses a fixed published ratio by trying finite h and anisotropy ladders and accepting only the measured matrix gates; that is an implementation policy, not a determination of a universal critical ratio. Close together with U1, in two dimensions first so that rung L2 provides a validated procedure.
U9	Whether the extrusion symmetry survives	The weight table presumes a node cloud that is a uniform extrusion along the homogeneous direction. Relaxing this in Stage 2 multiplies the table by the number of nodes in that direction. Close by deciding, before Stage 2, whether variable resolution is wanted in the homogeneous direction at all.

8 Interface with the existing solver

- A compile-time switch selects the present streaming path; it is mutually exclusive with the existing streaming paths, enforced at compile time.
- No existing streaming, collision or infrastructure module is modified.
- The collision operator, the forcing scheme, the statistics, the file input/output, the halo exchange, the mass correction and its monitoring are reused unchanged.
- **Ghost storage is retained, but it is not part of a LABFM cloud.** The incumbent wall boundary, ITB/GILBM streaming and statistics-window contracts still require their existing ghost layers and must not be dismantled. Independently, the P6/31 selector of §3.7 admits canonical physical IDs only and clamps its one-sided stencil into the domain. If a finite-difference hybrid in the sense of [2, Table 1] is adopted later, the interface between that strip and the meshless interior must be specified as a separate method change before any production run is submitted.
- The stencil is centred on the arrival node, which restricts the time step through Proposition 3. Any change to the Courant number must be checked against both that bound and the relaxation-time bound.
- The scheme is specified for a single global time step. Local time stepping is out of scope.

A Provenance table

This report	Source	Content
(1), (2)	[1, Eqs. (1),(2)], [2, Eqs. (1),(2)]	Monomial and derivative vectors
(3)	[1, Eq. (3)]	Compact Taylor expansion
(4)	[1, Eq. (4)], [2, Eq. (3)]	General discrete operator; $\mathbf{C}^d$ from [1, Eq. (5)], [2, Eq. (4)]
(5)	[1, Eq. (10)], [2, Eq. (5)]	ABF expansion of the weights
(6)	[1, Eqs. (13),(14)], [2, Eqs. (6),(7)]	Moment matrix and local system
(8)	[1, Eqs. (15),(16)]	Basis dimension in two and three dimensions
(11)	[1, Eq. (19)]	Original ABFs
(12)	[2, Eqs. (10),(11)]	Hermite ABFs
(13)	[2, Eq. (9)]	Row preconditioning by powers of h
§3.6 table	[1, §3.2], [2, §§4, 5]	Critical (original ABFs) and recommended (Hermite) stencil ratios; stencil optimisation
§3.10	[1, §4.1], [2, §7.1, Table 1, Eq. (24), Appendix A]	Boundary treatment; the 2021 retraction of the 2020 one-sided claim and the finite-difference hybrid
§3.7, (25)–(52)	Current project sources: <code>labfm_stencil.h</code> , <code>labfm_solve.h</code>	Two-dimensional P6/31-point candidate retention, mandatory template/QRCP membership, anisotropic scaling, rank and rcond gates, and compact-first repair selection; an implementation record, not a claim from the cited LABFM literature
(15)	[N]	Three-dimensional neighbour count; geometric extension of the two-dimensional relation in [2, §2]
(61)–(64)	[N]	Departure-point interpolation operator (Definition 1)
Theorem 1	[N]	Exact polynomial reproduction at the departure point
Corollaries 2, 3	[N]	Constant preservation; zero displacement identity
Remark 10	[N]	Approximating, not interpolating; invalidates a natural but incorrect regression test
(66)	[N]	Amplification constant of the interpolation operator
(67), (68)	[N], adapted from [2, §4, Eqs. (13a)–(13c)]	Von Neumann amplification and stability criterion for the streaming step
Theorem 4, (69)	[N]	Affine invariance; three dimensional Hermite ABFs
Remark 2, (7)	[N]	Distributional reading; $\mathbf{C}$ as the moment vector of a target distribution
Remark 5	[1, Appendix A]	Truncation of the original ABFs for $m \geq 4$, required to reproduce the published results
Remark 7, (14)	[N]	Normalisation of the sampling condition
(59), (60)	[N]	Parity structure of the continuum limit; the parity defect as a leading sampling indicator
(9), (10)	[N]	Underdetermination of the weight system; the selector formulation unifying both arms

continued on next page

This report	Source	Content (cont.)
Propositions 1, 2, Remarks 3, 4	[N]	Degenerate limit $N_i = p$; gauge freedom of the expansion origin and the two origins that are not free
Remark 9, Proposition 3, (62)	[N]	Roles of $\mathbf{r}_i, \mathbf{r}_j, \mathbf{r}_q^*$; Courant bound for a node-centred stencil
(70)	[6, 7, 8]	Moving least squares; node-centred (generalised) kernel variant
(71), Theorem 5	[N]	Departure-point form of the control operator; equivalence of polynomial reproduction
(72), (73), H1	[N]	Gram versus Petrov–Galerkin conditioning; the sole surviving discriminator
§§5.5, 5.6	[N]	Fairness conditions and decision rule for the head-to-head comparison

B A worked reading of [1, §3.1]

The construction of [1, §3.1] is short, and its brevity conceals what is being asserted. This appendix restates it in full, and then carries it out in closed form on the smallest non-trivial configuration, so that every claim can be checked by hand.

Precedence. This appendix is a restatement, not a source. Sections B.1–B.3 assert nothing that is not in [1, §§2, 3.1]; their correctness consists in being a faithful transcription, so they cannot serve as the authority for their own content. **Where this appendix and [1] disagree, [1] governs and this appendix is in error.** Only the worked example of §§B.4, B.8, Proposition 4 and the material marked [N] stand on their own, and they function as checks on the transcription rather than substitutes for the source. The reader is expected to return to [1] whenever rungs L1–L4 or L6 fail, whenever a parameter such as h/s or m is to be chosen, whenever the work is to be cited externally, and — above all — whenever a statement here is doubted.

The example also exhibits, in a setting where all quantities are explicit, three properties established abstractly in the main text: the cancellation of the basis functions in the degenerate case (Proposition 1), the parity structure of the moment matrix (59), and the necessity of a lower bound on h/s (§3.6).

B.1 The statement to be proved

Section 2 of [1] establishes an identity rather than an approximation. Substituting the truncated Taylor expansion (3) into the discrete operator (4) gives [1, Eq. (7)]

$$\mathcal{L}_i^d(\phi) = \underbrace{\left(\sum_j x_{ji} w_{ji}^d\right)}_{B_x^{d,1}} \frac{\partial \phi}{\partial x} + \underbrace{\left(\sum_j y_{ji} w_{ji}^d\right)}_{B_y^{d,1}} \frac{\partial \phi}{\partial y} + \underbrace{\left(\sum_j \frac{x_{ji}^2}{2} w_{ji}^d\right)}_{B_{xx}^{d,2}} \frac{\partial^2 \phi}{\partial x^2} + \cdots \quad (74)$$

The discrete operator is therefore *exactly* a linear combination of the derivatives at node i , in which the coefficient of each derivative is a moment of the weights. Approximating a prescribed derivative is consequently not a question of choosing a good formula but of solving an algebraic problem: set the moment belonging to the desired derivative to unity and every other moment to zero. The moments that are required to vanish are precisely the error terms of the title of [1, §3.1]; this is the sense in which they are “eliminated”, and the elimination is exact, not asymptotic.

Smoothed particle hydrodynamics obtains its weights from a kernel derivative [1, Eq. (9)] and therefore accepts whatever moments that kernel happens to possess; [1] observes that the result “is not particularly useful”, being of zero order. The contribution of [1, §3.1] is a constructive means of imposing the moments instead of inheriting them.

B.2 The construction

The obstruction is that the moment conditions are p equations in N_i unknowns and, as discussed in §3.3, are underdetermined whenever $N_i > p$. The device of [1, §3.1] is to *parameterise* the weights rather than solve for them: by (5) the N_i unknown weights are replaced by p unknown coefficients multiplying p prescribed functions of the separation. Substituting this parameterisation into the moment vector and extracting the coefficients from the sum produces a square system, (6). The two forms of that system carry opposite intent and should not be conflated: [1, Eq. (12)] maps a given Ψ to the moments it produces, whereas [1, Eq. (14)] prescribes the moments and solves for Ψ . Only the second is the method.

B.3 What the dimensions are free to be, and what fixes them

[1, §3.1] closes the construction with a sentence that is easy to read as bookkeeping:

“The above analysis imposes no constraint on the lengths of X , W , C , and the size of M , although for consistency between (13) and (14), we require X , W and C to be of equal length, and M square.”

It is not bookkeeping. Read carefully it separates a genuine design decision from two consequences of it, and it is worth unpacking because the freedom it does *not* remove is the freedom the rest of this report depends on.

The first clause is literally true. Nothing in (3)–(6) refers to a dimension. The moment expansion (74) has infinitely many terms; $\mathbf{X}$ and $\mathbf{W}$ are formally infinite sequences; the outer product is formally an infinite array. Truncation is *imposed* at this point in the argument, not derived from anything preceding it. That is the content of “imposes no constraint”.

Of the three stated equalities, only one is a choice. Write $n_X = |\mathbf{X}|$, $n_W = |\mathbf{W}|$, $n_C = |\mathbf{C}|$. Then $\mathbf{M}$ has shape $n_X \times n_W$, so $\mathbf{M}\Psi$ lies in $\mathbb{R}^{n_X}$, and (6) equates it to $\mathbf{C}$. Hence

$$n_C = n_X \quad \text{is forced by the equation being well formed,} \quad (75)$$

and is not a modelling decision at all. The only genuine choice is n_W against n_X , and setting $n_W = n_X \equiv p$ is what makes $\mathbf{M}$ square. The three lengths carry three different meanings, and it is their coincidence that is the design decision:

Vector	Its length counts	Status
$\mathbf{X}$	the moment conditions imposed, i.e. the error terms annihilated	sets p
$\mathbf{W}$	the free coefficients available in Ψ	the choice
$\mathbf{C}$	the values prescribed	forced, (75)

In words: *one free coefficient per moment condition*. The squareness of $\mathbf{M}$ is the consequence, not the motive.

Proposition 4 (The two alternatives; [N]). *Let $n_X = p$ and let $\mathbf{M}$ have full rank. Then*

- (i) if $n_W > p$, the system (6) is underdetermined, its solution set is an affine subspace of dimension $n_W - p$, and every member satisfies all p moment conditions; polynomial reproduction, and hence the order, is unaffected, but a further rule is required to select Ψ ;
- (ii) if $n_W < p$, the system is overdetermined and generically inconsistent; no Ψ satisfies all p conditions, the unsatisfied moments survive in (74), and the order falls.

Proof. Immediate from the shape $p \times n_W$ of $\mathbf{M}$ and the rank hypothesis. For (i), membership of the solution set means $\mathbf{M}\Psi = \mathbf{C}$ holds exactly, which is the definition of the moment conditions; Theorem 1 then applies verbatim. For (ii), a least-squares surrogate leaves a non-zero residual in at least one moment, and that moment multiplies its derivative in (74). $\square$

Remark 14 (n_W interpolates between two extremes). *Case (i) is not pathological; it is a family of admissible schemes that [1] declines to enter. Its extreme is instructive. Since $\mathbf{w} = \mathbf{W}\Psi$, the reachable weights lie in the column space of $\mathbf{W}$, of dimension n_W . At $n_W = p$ the ansatz determines $\mathbf{w}$ completely once $\mathbf{C}$ is given; at the opposite extreme $n_W = N_i$ with $\mathbf{W}$ invertible, the reachable set is all of $\mathbb{R}^{N_i}$ and the ansatz imposes nothing, returning the bare underdetermined system (9). The choice $n_W = p$ therefore places LABFM at the minimal end of that range: the smallest parameterisation that still closes the system. Every intermediate value would introduce a second layer of arbitrariness on top of the selection already discussed in §3.3.*

Square is necessary, not sufficient. The quoted sentence secures only the shape of $\mathbf{M}$. Its invertibility is a separate matter and is where the method actually fails in practice: through the sampling condition of §3.6, and, in three dimensions with the original basis, through the unconditional rank deficiency of §3.4.

A second restriction, which the sentence does not state. Equality of the three lengths does not by itself determine p ; [1, Eqs. (15),(16)] do, and they admit only the values 2, 5, 9, 14, ... in two dimensions and 3, 9, 19, 34, 55, 83 in three. The reason is that p must fall at the end of a complete degree. Truncating in mid-degree — taking, in two dimensions, $\mathbf{X} = [x, y, x^2/2]$ with $p = 3$ — controls B_x , B_y and B_{xx} while leaving B_{xy} and B_{yy} free. Since $B^{(n)} \sim h^{n-l}$, those surviving second moments contribute at $O(h)$ to a first derivative where a complete truncation would have left $O(h^2)$: the order drops, and it drops *anisotropically*, a field varying along x retaining an accuracy that a field varying along y does not. Equivalently, $\text{span}\{1, x, y, x^2\}$ is not invariant under rotation of the axes whereas $\mathbb{P}_2$ is, so the operator would depend on an orientation that is arbitrary on a body-fitted curvilinear grid. This is enforced structurally in the implementation of §3.4: the multi-index enumeration loops over complete degrees, so a mid-degree truncation cannot be expressed.

What remains free. The sentence locks the ansatz dimension to the constraint count. It says nothing about N_i , and N_i is not locked: the necessary condition is only $N_i > p$, and the surplus $N_i - p$ is precisely the freedom exploited in §3.3 and contested between the two arms of §5. The division is therefore sharp, and worth stating as the summary of this subsection: *the number of unknowns in the parameterisation is fixed by the number of conditions; the number of data points is not.*

B.4 A minimal example

Let the problem be one-dimensional with $m = 2$, so that $\mathbf{X}_{ji} = [x_{ji}, x_{ji}^2/2]^\top$ and $p = m = 2$; let node i lie at $x = 0$ and let its neighbours lie at $x = \pm s$. Then

$$\mathbf{X}_{j1i} = \left[s, \frac{s^2}{2} \right]^\top, \quad \mathbf{X}_{j2i} = \left[-s, \frac{s^2}{2} \right]^\top, \quad \det \mathbf{X} = s^3 \neq 0, \quad (76)$$

so the moment conditions $\mathbf{X}^\top \mathbf{w} = \mathbf{C}$ possess a unique solution for every right-hand side. Written out, they are

$$s(w_1 - w_2) = C_1, \quad \frac{s^2}{2}(w_1 + w_2) = C_2. \quad (77)$$

First derivative. Setting $\mathbf{C} = [1, 0]^\top$, the second of (77) forces $w_2 = -w_1$ — this is the elimination of the ϕ'' error term — whence $w_1 = 1/2s$ and $w_2 = -1/2s$, and

$$\mathcal{L}_i^x(\phi) = \frac{\phi_{j_1} - \phi_i}{2s} - \frac{\phi_{j_2} - \phi_i}{2s} = \frac{\phi(s) - \phi(-s)}{2s}. \quad (78)$$

Second derivative. Setting $\mathbf{C} = [0, 1]^\top$, the first of (77) forces $w_1 = w_2$ — the elimination of the ϕ' error term — whence $w_1 = w_2 = 1/s^2$, and

$$\mathcal{L}_i^L(\phi) = \frac{\phi_{j_1} - \phi_i}{s^2} + \frac{\phi_{j_2} - \phi_i}{s^2} = \frac{\phi(s) - 2\phi(0) + \phi(-s)}{s^2}. \quad (79)$$

The machinery of [1, §3.1] thus returns the classical central and second differences. This is the expected outcome and is worth stating explicitly, because it identifies what the method generalises: LABFM is a procedure for constructing, on an arbitrary node distribution, the object that the moment conditions define uniquely on a uniform one.

B.5 The superscript d : what it does and does not affect

The label d appears on w_{ji}^d , Ψ_i^d , $\mathbf{B}_i^d$ and $\mathbf{C}^d$ but not on $\mathbf{X}_{ji}$, $\mathbf{W}_{ji}$ or $\mathbf{M}_i$. That asymmetry is the whole content of the notation, and it is worth making explicit, because the extension of §4 consists of nothing but replacing d by a different label.

Dependency chain. Reading (6) and (5) from right to left,

$$\mathbf{C}^d \mapsto \Psi_i^d = \mathbf{M}_i^{-1} \mathbf{C}^d \mapsto w_{ji}^d = \mathbf{W}_{ji} \cdot \Psi_i^d \mapsto \mathbf{B}_i^d = \sum_j \mathbf{X}_{ji} w_{ji}^d (= \mathbf{C}^d). \quad (80)$$

Only $\mathbf{C}^d$ is supplied; the remaining three are consequences. The superscript is therefore *passive*: it enters no formula except as the argument of the solve, and it is carried on Ψ , w and $\mathbf{B}$ only because those objects inherit the dependence. In particular $\mathbf{B}_i^d$ is the vector of moments *achieved*, and after the solve it coincides with the vector *prescribed*; [1] writes $\mathbf{B}$ in Eq. (8), before the ansatz, when it is whatever the weights happen to give, and replaces it by $\mathbf{C}$ in Eq. (14), when it is demanded.

The passivity is not a redundancy, however: for each d the objects Ψ_i^d and w_{ji}^d are genuinely different arrays of numbers, as the example below shows.

A worked example in two dimensions. Take $m = 2$, so that $p = 5$ and

$$\mathbf{X}_{ji} = \left[x_{ji}, y_{ji}, \frac{x_{ji}^2}{2}, x_{ji}y_{ji}, \frac{y_{ji}^2}{2} \right]^\top,$$

and place five neighbours at

$$(\pm s, 0), \quad (0, \pm s), \quad (s, s), \quad (81)$$

numbered $j_1, \dots, j_5$ in that order, so that $N_i = p$ and, by Proposition 1, the basis functions cancel and the weights are those of the moment conditions alone. Writing out $\mathbf{X}^\top \mathbf{w} = \mathbf{C}$ row by row,

$$\begin{aligned} s(w_1 - w_2) + s w_5 &= C_1, & s(w_3 - w_4) + s w_5 &= C_2, & \frac{s^2}{2}(w_1 + w_2 + w_5) &= C_3, \\ s^2 w_5 &= C_4, & \frac{s^2}{2}(w_3 + w_4 + w_5) &= C_5. \end{aligned} \quad (82)$$

Solving for the three standard right-hand sides of [1, Eq. (5)]:

d	$\mathbf{C}^d$	$\mathbf{w}^d = (w_1, \dots, w_5)$	resulting operator
x	$[1, 0, 0, 0, 0]$	$[\frac{1}{2s}, -\frac{1}{2s}, 0, 0, 0]$	$\frac{\phi_E - \phi_W}{2s}$
y	$[0, 1, 0, 0, 0]$	$[0, 0, \frac{1}{2s}, -\frac{1}{2s}, 0]$	$\frac{\phi_N - \phi_S}{2s}$
L	$[0, 0, 1, 0, 1]$	$[\frac{1}{s^2}, \frac{1}{s^2}, \frac{1}{s^2}, \frac{1}{s^2}, 0]$	$\frac{\phi_E + \phi_W + \phi_N + \phi_S - 4\phi_i}{s^2}$

The three classical formulae are recovered, from one and the same $\mathbf{M}_i$, by changing $\mathbf{C}^d$ alone. Three features of the table answer the question of what the superscript does.

- (i) *The weights are different arrays, not a relabelling.* $\mathbf{w}^x$ is supported on the two horizontal neighbours, $\mathbf{w}^y$ on the two vertical ones, and $\mathbf{w}^L$ on all four; the fifth neighbour at (s, s) receives zero weight in all three cases, since $C_4 = 0$ throughout and (82) gives $w_5 = C_4/s^2$.
- (ii) *The superscript carries a dimension.* $w^x \sim s^{-1}$ while $w^L \sim s^{-2}$. This is the scaling requirement of [1, §2], that w^d scale as h^{-l} with l the order of the derivative; for the interpolation operator of §4, $l = 0$ and the weights are dimensionless, which is why the affine rescaling of §4.5 requires no inverse transformation.
- (iii) *The dependence on $\mathbf{C}^d$ is linear.* Since $\Psi_i^d = \mathbf{M}_i^{-1} \mathbf{C}^d$ and $w^d = \mathbf{W} \Psi^d$, the map $\mathbf{C}^d \mapsto \mathbf{w}^d$ is linear. Hence operators superpose: with $\mathbf{C}^{xx} = [0, 0, 1, 0, 0]$ and $\mathbf{C}^{yy} = [0, 0, 0, 0, 1]$ one has $\mathbf{C}^L = \mathbf{C}^{xx} + \mathbf{C}^{yy}$ and therefore $\mathbf{w}^L = \mathbf{w}^{xx} + \mathbf{w}^{yy}$, which the table verifies term by term. The hyperviscosity operators of [1, Eq. (25)] are constructed in exactly this way, and so is any other linear combination of derivatives one may require.

Remark 15 (Why this is the entry point for the present work). *Since d enters only through $\mathbf{C}^d$, and $\mathbf{M}_i$ is indifferent to it, the label may be replaced by any index whatever provided a corresponding right-hand side is supplied. §4 replaces $d \in \{x, y, L, \dots\}$ by the discrete velocity $q \in \{0, \dots, 18\}$ and the right-hand side by $\mathbf{X}(\delta_q)$. Nothing else in the construction changes, which is why one factorisation continues to serve every case (Remark 1). The passivity of the superscript is precisely what makes the extension available.*

B.6 The same computation through the basis functions

The example above bypassed (5). Carrying the parameterisation through exhibits its role. Adopt the Hermite construction in the probabilists' normalisation of §3.4, write $a \equiv s/h_i$ and $\psi_a \equiv \psi(a)$, and use $\text{He}_1(z) = z$, $\text{He}_2(z) = z^2 - 1$.

Here $p = 2$ and $N_i = 2$, so there are four numbers W_{ji}^α , one for each pair of a basis function and a neighbour. It is worth displaying them as an array, since the roles of the two indices are the point of the exercise (Remark 6): *both* basis functions are evaluated at *both* neighbours, and differ between them only through the argument.

	basis $\alpha=1$: $\text{He}_1(z) = z$	basis $\alpha=2$: $\text{He}_2(z) = z^2 - 1$
neighbour j_1 ($\tilde{x} = +a$)	$\psi_a a$	$\psi_a (a^2 - 1)$
neighbour j_2 ($\tilde{x} = -a$)	$-\psi_a a$	$\psi_a (a^2 - 1)$

The rows of this array are the vectors of (5),

$$\mathbf{W}_{j_1 i} = \psi_a [a, a^2 - 1]^\top, \quad \mathbf{W}_{j_2 i} = \psi_a [-a, a^2 - 1]^\top, \quad (83)$$

and the array itself is the matrix $\mathbf{W}$ of §3.3, of shape $N_i \times p$. The sign difference between the rows in the first column, and its absence in the second, is the parity of He_1 and He_2 ; it is what produces the diagonal structure below.

Assembling $\mathbf{M}_i = \sum_j \mathbf{X}_{ji} \otimes \mathbf{W}_{ji}$ entrywise — each entry being a sum over neighbours, its position fixed by (which monomial, which basis function) —

$$M_{11} = \underbrace{s \cdot \psi_a a}_{\text{from } j_1} + \underbrace{(-s) \cdot (-\psi_a a)}_{\text{from } j_2} = 2sa\psi_a, \quad M_{12} = \underbrace{s \cdot \psi_a(a^2-1)}_{\text{from } j_1} + \underbrace{(-s) \cdot \psi_a(a^2-1)}_{\text{from } j_2} = 0, \quad (84)$$

and likewise $M_{21} = 0$ and $M_{22} = s^2\psi_a(a^2 - 1)$, so that

$$\mathbf{M}_i = \mathbf{X}^\top \mathbf{W} = \begin{pmatrix} 2sa\psi_a & 0 \\ 0 & s^2\psi_a(a^2 - 1) \end{pmatrix}. \quad (85)$$

The off-diagonal entries vanish identically, in agreement with (59): the stencil is symmetric, and an entry pairing a monomial of one parity with a Hermite function of the other integrates to zero. Solving (6) for $\mathbf{C} = [1, 0]^\top$ gives $\Psi = [(2sa\psi_a)^{-1}, 0]^\top$ and, through (5),

$$w_{j_1} = \psi_a a \cdot \frac{1}{2sa\psi_a} = \frac{1}{2s}, \quad w_{j_2} = -\psi_a a \cdot \frac{1}{2sa\psi_a} = -\frac{1}{2s}, \quad (86)$$

identical to (78); the same cancellation occurs for $\mathbf{C} = [0, 1]^\top$. Neither ψ nor a survives. This is Proposition 1 in the smallest possible instance: with $N_i = p$ the parameterisation is a bijection, the selection it performs is vacuous, and the choice of basis is immaterial.

B.7 A degeneracy of the parameterisation that the problem does not possess

Equation (85) is singular when $a = 1$, that is when $h_i = s$, since the second diagonal entry then vanishes. At that ratio the second-order operator cannot be constructed through the ansatz, although the underlying moment conditions (77) remain uniquely solvable, $\mathbf{X}$ being invertible by (76). The failure therefore belongs to the parameterisation and not to the problem.

This is the mechanism behind the requirement of §3.6 in its simplest form. The statement of [1, §3.2] that the node distribution must “adequately sample the ABFs” is, in this example, the statement that h_i/s must be kept away from a value at which the chosen basis becomes blind to a moment it is required to control. It is consistent that the operating ratios recommended in [2, §5] for $m = 2$ lie above unity; the example does not derive that value, but it does show why some such condition is unavoidable, and why the condition attaches to the basis rather than to the accuracy requirement.

B.8 The underdetermined case

Retain $m = 2$ and add a third neighbour at $x = 2s$, so that $N_i = 3 > p = 2$. The moment conditions for $\mathbf{C} = [1, 0]^\top$ read

$$s(w_1 - w_2 + 2w_3) = 1, \quad \frac{s^2}{2}(w_1 + w_2 + 4w_3) = 0, \quad (87)$$

whose general solution is the one-parameter family

$$w_3 = t, \quad w_1 = \frac{1}{2s} - 3t, \quad w_2 = -\frac{1}{2s} - t, \quad t \in \mathbb{R}. \quad (88)$$

Every member reproduces polynomials of degree at most two exactly: on $\phi = x$ the operator returns $s(w_1 - w_2 + 2w_3) = 1$, and on $\phi = x^2$ it returns $s^2(w_1 + w_2 + 4w_3) = 0$, independently of t . The choice $t = 0$ recovers (78); $t = 1/6s$ gives the equally consistent formula $(0, -2/3s, 1/6s)$.

The members are not, however, indistinguishable. Applying the operator to $\phi = x^3$, the lowest degree not controlled by the moment conditions, gives

$$\mathcal{L}_i^x(x^3) = s^3(w_1 - w_2 + 8w_3) = s^2 + 6s^3t, \quad (89)$$

against an exact value of zero. The error is $O(s^2) = O(s^m)$ for every t , as the order statement requires, but its coefficient depends linearly on t : the family shares one order of accuracy and not one error constant. The precise statement is therefore that the surplus degrees of freedom cannot change the *order*, only the coefficient of the leading error term, together with the conditioning and stability properties of the resulting operator. It is this coefficient, and not the order, that the resolving-power optimisations of [3, 4] address.

Accuracy of a given order therefore does not distinguish the members of (88), and some further principle must. In LABFM that principle is the restriction of $\mathbf{w}$ to $\text{span}(\mathbf{W})$, which selects one value of t ; in the control formulation of §5 it is a weighted least-squares minimisation, which selects another. The surplus degree of freedom is spent on conditioning and stability, as recorded in Remark 3, and the requirement $N_i > p$ is not optional: the critical neighbour counts of [1, §3.2] exceed the corresponding polynomial counts at every order.

B.9 The remaining assertions of [1, §3.1]

Four statements close the section, and each may now be read directly.

- (i) *Only the first p elements of $\mathbf{X}$ and $\mathbf{W}$ are retained.* The expansion (74) has infinitely many terms; finitely many can be annihilated. The truncation point defines the method’s order and fixes the size of $\mathbf{M}_i$ through (8).
- (ii) *The leading error scales as h^{m+1-l} .* The first moment *not* prescribed is that of total degree $m + 1$, and it multiplies a derivative of that order; dividing by the h^{-l} scaling of the weights for an l -th derivative gives the quoted exponent. For $l = 0$, the case of interest here, the exponent is $m + 1$.
- (iii) *$\mathbf{M}_i$ and $\mathbf{w}^d$ depend on the anisotropy of the distribution, yet the accuracy is maintained.* The diagonal form (85) is an artefact of symmetry; on a disordered cloud $\mathbf{M}_i$ is full, and its off-diagonal entries measure the extent to which each basis function contributes to moments other than its own. That contamination is inverted by solving (6), which is why the accuracy is unaffected while the conditioning is not.
- (iv) *$\mathbf{M}_i$ is the same for every operator.* The matrix is assembled from the stencil geometry alone; the operator enters only through the right-hand side. The present work rests on this observation, and §4 exploits it by supplying a right-hand side that no longer corresponds to a derivative.

B.10 Relation to the operator of §4

In the language of Remark 2, [1, §3.1] specifies the moments of a derivative of a Dirac distribution centred at $\mathbf{r}_i$, and the entries of $\mathbf{C}^d$ in [1, Eq. (5)] are exactly those moments. The operator of Definition 1 changes nothing in the construction except the target: the moments of a Dirac distribution translated to $\mathbf{r}_i + \boldsymbol{\delta}_q$, which are the monomials evaluated at the displacement. Where [1, §3.1] sets almost every moment to zero, the present right-hand side prescribes all of them to non-zero values; the word “elimination” is then no longer descriptive, but the counting argument that determines the order — the first unprescribed moment is the leading error — is unchanged, and is what Theorem 1 formalises.

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